

Dynamic Machine Learning Modelling of Longitudinal Multi-Omics for Predicting Long-Term Weight Loss Maintenance

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ABSTRACT

Long-term weight loss maintenance remains a central challenge in metabolic health, as most individuals regain weight following dietary interventions. While longitudinal multi-omics profiling provides detailed biological measurements over time, applying machine learning to these datasets is severely constrained by the curse of dimensionality, where the number of measured biological variables vastly exceeds the available clinical sample size ($p \gg n$).

Using data from the Finnish ELIPA study ($N = 82$, tracked across baseline, post-diet, and 6-month maintenance timepoints), this dissertation models dynamic rates of biological change (velocity features, $\Delta\text{value}/\Delta t$) across acute weight loss (WL) and weight loss maintenance (WM) phases. Exploratory analysis revealed a strong *Phase Independence* effect: 98.4% of metabolomic velocity pairs between the weight loss and maintenance phases were negatively correlated (mean $r = -0.54$). This indicates a physiological rebound where the acute response to caloric restriction does not directly predict long-term maintenance success.

Seven machine learning architectures spanning flat models (CatBoost, Random Forest, ElasticNet, Multi-Layer Perceptron) and sequential networks (GRU, LSTM, Temporal Self-Attention) were systematically benchmarked under leak-free 5-fold cross-validation. On metabolomics data (Dataset B, $N = 79, P = 199$ features), CatBoost achieved the strongest baseline performance ($R^2 = 0.223$), improving to $R^2 = 0.384$ when pruned to the top 50 features. On transcriptomics data (Dataset C, $N = 31, P = 200$ features, filtered from 39,250 raw gene velocities), the Multi-Layer Perceptron achieved the highest performance ($R^2 = 0.529$), followed by the GRU ($R^2 = 0.500$) and CatBoost ($R^2 = 0.480$).

Directly concatenating all features into a single early fusion dataset (Dataset A, $N = 33, P = 1,169$) resulted in severe overfitting with negative R^2 scores across all models. To overcome this limitation, a modular Multi-Modal Late Decision Fusion pipeline was developed on the paired cohort ($N = 30$). By training separate models on each omic modality and combining their out-of-fold predictions using regularised Ridge Stacking, the late fusion model achieved the highest overall accuracy ($R^2 = 0.576$, RMSE = 3.014%, MAE = 2.220%), outperforming single modalities, simple averaging ($R^2 = 0.376$), and non-linear decision tree stacking ($R^2 = 0.438$).

Explainable AI analysis using SHAP values and Spearman rank correlations confirmed strong cross-architecture consensus (ρ up to 0.623, $p < 10^{-20}$), identifying Indolelactate in metabolomics and PAOX in transcriptomics as top predictive biomarkers across different model families. In addition, the Temporal Self-Attention network independently allocated 70–80% of its attention weight to the maintenance phase, and permutation testing verified that predictive performance was statistically significant ($p < 0.001$).

For the ELIPA study, these findings show that weight maintenance outcomes are largely decoupled from acute weight loss and are primarily driven by biological stabilisation during the maintenance period. This advances the trial’s characterisation from single-modality correlations to integrated predictive biomarkers. Generally, Regularised late fusion and velocity-based modelling are effective for the curse of dimensionality in small clinical cohorts. The discovery of phase independence and cross-model consensus biomarkers confirms that maintenance steady-state adaptations govern long-term stability, providing actionable targets for precision nutrition.

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I am deeply grateful to Tingyu Guo for generously providing the ELIPA clinical multi-omics dataset and sharing insights from her foundational PhD research. Her prior benchmarking work served as an essential baseline and inspiration for the machine learning models and multi-modal integration frameworks developed in this thesis.

Finally, I would like to thank the participants and clinical research staff of the Finnish ELIPA study, whose dedicated involvement made the collection of this longitudinal dataset possible.

ABBREVIATIONS AND GLOSSARY

List of Abbreviations

Abbrev.	Description	Abbrev.	Description
AhR	Aryl Hydrocarbon Receptor	MAE	Mean Absolute Error
ARI	Adjusted Rand Index	MDI	Mean Decrease in Impurity
BMI	Body Mass Index	MLP	Multi-Layer Perceptron
CAR	Acylcarnitine	MW	Molecular Weight
CB	CatBoost	O- / P-	Ether / Plasmalogen lipid
CV	Cross-Validation	OOF	Out-Of-Fold
ELIPA	ELIPA Study	PAOX	Polyamine Oxidase
EN	ElasticNet	PC	Phosphatidylcholine
GBDT	Gradient Boosted Trees	PCA	Principal Component Analysis
GRU	Gated Recurrent Unit	RF	Random Forest
ILA	Indole-3-Lactic Acid	RMSE	Root Mean Square Error
LCD	Low-Calorie Diet	R^2	Coefficient of Determination
LOD	Limit of Detection	SHAP	SHapley Additive exPlanations
LPC	Lysophosphatidylcholine	SSAT	Spermidine/Spermine Acetyltransferase
LPE	Lysophosphatidylethanolamine	VLCD	Very-Low-Calorie Diet
LSTM	Long Short-Term Memory	WL / WM	Weight Loss / Maintenance
MKL	Multiple Kernel Learning	XAI	Explainable Artificial Intelligence

Lipid and Metabolomic Nomenclature

Lipid identifiers (e.g., **PC 38:7_67**, **LPC(17:0)**) follow standard shorthand conventions:

- **Headgroup / Class:** Core phospholipid category (**PC** = Phosphatidylcholine, **LPC** = Lysophosphatidylcholine, **CAR** = Acylcarnitine).
- **Carbon Count & Double Bonds:** In **PC 38:7**, **38** denotes total aliphatic carbons, and **7** denotes double bonds.
- **Isomer Suffix:** Suffixes (**_67**, **_68**) distinguish chromatographic isomers resolved on LC-MS.
- **Linkage Prefixes:** **O-** indicates an alkyl ether lipid; **P-** indicates a vinyl ether plasmalogen.

Gene and Transcriptomic Nomenclature

Gene symbols follow HGNC guidelines and designate regulatory targets in adipose transcriptomics:

- **PAOX (Polyamine Oxidase):** Peroxisomal enzyme oxidizing N^1 -acetylated polyamines, driving an energy-consuming futile metabolic cycle in white adipose tissue.
- **TSPAN12 (Tetraspanin 12):** Endothelial transmembrane coreceptor regulating Norrin-FZD4-LRP5- β -catenin signaling in adipose microvascular remodeling.
- **KLF15 (Krüppel-like Factor 15):** Transcriptional master regulator of branched-chain amino acid (BCAA) catabolism and gluconeogenesis in adipose depots.
- **RHAG (Rh-associated Glycoprotein):** Transmembrane channel facilitating cellular ammonium (NH_4^+) and CO_2 clearance during catabolic stress.
- **EPM2A (Laforin Glycogen Phosphatase):** Dual-specificity phosphatase essential for glycogen homeostasis and cellular proteostasis.

Biochemical Pathways and Microbial Metabolites

- **Indolelactate (Indole-3-lactic acid, ILA):** Commensal gut microbiota-derived tryptophan catabolite acting as an endogenous agonist for the Aryl Hydrocarbon Receptor (AhR), fortifying gut epithelial barrier integrity.
- **Kynurenine Pathway:** Host-mediated tryptophan degradation cascade induced by interferon- γ and IDO1, serving as a clinical marker of systemic metabolic inflammation.
- **Odd-Chain Fatty Acids (OCFAs):** Fatty acids with odd carbon counts (e.g., C17:0 in LPC(17:0)), synthesized via gut microbial propionate or dairy intake, linked to metabolic health and insulin sensitivity.

Temporal Velocity Feature Notation

Longitudinal features engineered from clinical timepoints (t_1 = Baseline, t_2 = Post-WL, t_4 = Post-WM) are designated by explicit phase prefixes:

- **WL_vel_<feature>:** Rate of change during acute Weight Loss: $\frac{x(t_2) - x(t_1)}{\Delta t_{\text{WL}}}$.
- **WM_vel_<feature>:** Rate of change during Weight Maintenance: $\frac{x(t_4) - x(t_2)}{\Delta t_{\text{WM}}}$.

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1 INTRODUCTION

Obesity represents one of the most critical global public health challenges, affecting over 650 million adults worldwide and serving as a major causal driver of type 2 diabetes, cardiovascular diseases, musculoskeletal disorders, and metabolic syndrome [2, 21]. While clinical dietary interventions, such as Very-Low-Calorie Diets (VLCD), reliably produce substantial short-term weight reduction, long-term weight loss maintenance (WLM) remains remarkably difficult to sustain. Coordinated neuroendocrine counter-regulation (sustained suppression of leptin and elevation of orexigenic ghrelin) and adaptive thermogenesis drive weight regain in 80% to 90% of individuals within three to five years following an intervention [39, 16].

Identifying the molecular and physiological mechanisms that differentiate individuals who successfully sustain weight loss from those who experience weight regain is essential for developing personalized, precision nutrition interventions. However, static baseline clinical measurements (such as fasting glucose, lipid panels, or baseline BMI) provide limited predictive power regarding dynamic adaptive responses, as they represent late-stage manifestations rather than dynamic indicators of cellular flexibility [25, 6]. Human metabolism during dietary interventions is an evolving biological process characterized by complex neuroendocrine, transcriptomic, metabolomic, and gut microbial feedback loops.

Recent advancements in high-throughput multi-omics profiling offer unprecedented opportunities to observe these dynamic biological adaptations across multiple regulatory layers [31, 33]. Nevertheless, integrating high-dimensional, longitudinal multi-omics data into predictive machine learning frameworks introduces severe methodological hurdles. In clinical cohorts, the number of measured molecular features (p) frequently reaches tens of thousands (e.g., in transcriptomic arrays), while the number of enrolled human participants (n) is severely constrained ($n \approx 30\text{--}80$). This extreme $p \gg n$ regime, coupled with complex missingness, temporal phase transitions, and cross-platform heterogeneity, creates an acute risk of overfitting and spurious biomarker discovery.

1.1 Research Context and the ELIPA Study

This dissertation investigates dynamic machine learning modelling using longitudinal multi-omics data collected during the Finnish ELIPA (Early Life and Interventions in Obesity Prevention and Adult Health) food intervention study. The ELIPA trial tracked participants with severe obesity ($\text{BMI} > 40 \text{ kg/m}^2$) through a multi-stage protocol comprising:

1. An initial 7-to-8 week rapid weight loss (WL) phase induced by a Very-Low-Calorie Diet (VLCD),
2. A 2-week transitional refeeding period, and
3. A 24-week (6-month) ad libitum weight maintenance (WM) phase.

Multi-omics measurements (including targeted and non-targeted metabolomics, gut microbiota profiling, whole-transcriptome microarray profiling, and clinical anthropometrics) were gathered at three primary timepoints: baseline (t_1), post-weight loss (t_2), and post-weight maintenance (t_4).

In preliminary work by Guo et al. [15], manual feature engineering was utilized to compute “velocity” features (representing the rate of change $\Delta\text{value}/\Delta t$ between consecutive clinical timepoints). Using a Random Forest regressor, Guo et al. demonstrated that dynamic velocity features could predict percentage weight change during maintenance and identified several candidate predictors, such as fat mass velocity and tryptophan-related metabolites.

However, several fundamental computational and biological questions remain unanswered:

- **Flat vs. Sequential Paradigms:** Can modern sequential deep learning models (such as LSTMs, GRUs, or Temporal Self-Attention networks) capture longitudinal trajectories more effectively than static “flat” tree ensembles or regularized linear models?
- **Temporal Coupling vs. Phase Independence:** Does an individual’s biological response during acute, rapid caloric restriction ($t_1 \rightarrow t_2$) directly forecast their ability to sustain weight loss during ad libitum maintenance ($t_2 \rightarrow t_4$), or are these phases biologically decoupled as observed in clinical intervention trials [44, 41]?
- **Dimensionality Bottlenecks and Feature Selection:** In high-dimensional transcriptomics ($P = 39,250$), where supervised Random Forest feature selection is required by default to make modeling tractable ($P = 200$), does applying the same RF feature selection strategy to lower-dimensional metabolomics ($P = 199$) yield comparable performance gains?
- **Multimodal Decision Fusion:** How can disparate omics modalities with mismatched sample sizes and feature dimensions be optimally combined without triggering feature explosion or catastrophic overfitting?
- **Consensus Interpretability:** How consistent are the biological biomarkers extracted by disparate model families, and can non-parametric rank correlation metrics (Spearman rank correlation) formally establish algorithmic consensus?

1.2 Project Objectives

To address these questions, this dissertation pursues five primary objectives:

1. **Longitudinal Feature Preprocessing and Dimensionality Management:** Design a leak-free preprocessing and feature engineering pipeline to compute multi-phase velocity features, prevent target leakage, and establish a tractable transcriptomic benchmark (Dataset C) using default Random Forest feature selection.
2. **Systematic Multi-Architecture Benchmarking:** Evaluate and compare four flat models (Random Forest, CatBoost, ElasticNet, Multi-Layer Perceptron) and three sequential deep learning models (LSTM, GRU, Temporal Self-Attention) under strict 5-fold cross-validation.
3. **Investigation of Phase Independence:** Statistically quantify the correlation between the rapid weight loss (WL) and weight maintenance (WM) phases across multi-omic layers to test the hypothesis of biological phase independence.
4. **Feature Selection and Late Fusion Ensembling:** Systematically test the impact of Random Forest feature selection on Dataset B (Metabolomics) across model families, and develop a decision-level multi-modal Late Fusion pipeline using regularized Ridge Stacking on the multi-omic intersection cohort.
5. **Explainable AI and Cross-Model Consensus Validation:** Extract mechanistic biological insights using SHapley Additive exPlanations (SHAP) and attention weights, and formally evaluate biomarker reliability across distinct algorithmic families using Spearman rank correlation analysis.

1.3 Key Contributions and Achievements

The primary contributions and scientific findings of this dissertation include:

- **Discovery and Quantification of Phase Independence:** Discovered that 98.4% of metabolomic velocity feature pairs between the WL and WM phases exhibit negative correlations (mean $r = -0.54$). This establishes that the acute weight loss phase represents a transient physiological perturbation followed by a homeostatic rebound, providing a clear mathematical explanation for why complex sequential models (LSTMs, GRUs) do not outperform flat models in this regime.
- **Exhaustive Algorithmic Benchmarking:** Demonstrated that gradient-boosted trees (CatBoost, $R^2 = 0.223$, rising to 0.384 under feature selection) and deep neural architectures (MLP, $R^2 =$

0.529 on transcriptomics) deliver strong predictive performance, outperforming traditional linear and un-boosted tree baselines.

- **Resolution of Transcriptomic Dimensionality and Feature Selection Analysis:** Established that while Random Forest feature selection is mandatory by default to reduce transcriptomics from 39,250 velocity features to 200 ($N = 31$), applying the same RF feature selection to metabolomics (Dataset B) substantially boosts flat models (e.g., CatBoost R^2 increasing by 72%) while severely degrading sequential models due to broken temporal pairings.
- **Superiority of Regularized Late Fusion:** Implemented a decision-level Ridge Stacking late fusion architecture on the multi-omic intersection cohort ($N = 30$) that achieved an R^2 of 0.576 and an RMSE of 3.014%, outperforming single-modality models and unregularized meta-learners.
- **Rigorous Cross-Model Consensus via Spearman Rank Correlation:** Validated that feature importance rankings across diverse algorithmic families (e.g., CatBoost vs. Random Forest $\rho = 0.527$ on Dataset B, $\rho = 0.524$ on Dataset C; ElasticNet vs. MLP $\rho = 0.623$ on Dataset C) exhibit statistically significant non-parametric agreement, establishing gut-microbiota-derived **Indolelactate** (tryptophan-AhR pathway) and cellular **PAOX** (polyamine catabolism) as robust, model-agnostic biomarkers of weight maintenance success.

1.4 Dissertation Structure

The remainder of this dissertation is organized into six chapters:

- **Chapter 2 (Background Theory and Literature Review):** Reviews the biological mechanisms of weight regulation, outlines the taxonomy of multi-omics data fusion (early, intermediate, late), discusses the statistical challenges of $p \gg n$ data, and details the mathematical foundations of flat, sequential, and explainable AI models.
- **Chapter 3 (Data Exploration & Phase Independence):** Details data cleaning, velocity feature engineering, target leakage prevention, missing value imputation, the default Random Forest feature selection for Dataset C, and provides quantitative empirical evidence for phase independence.
- **Chapter 4 (Modelling & Multi-Architecture Comparison):** Evaluates the seven machine learning architectures across Datasets B and C, presents the feature selection experiment on Dataset B, and details the multimodal Ridge Stacking late fusion pipeline.

- **Chapter 5 (Interpretability & Mechanistic Insights):** Applies SHAP and temporal attention weights to uncover biological mechanisms, explores the tryptophan-kynurenine and polyamine catabolic pathways, and contrasts these findings with prior literature.
- **Chapter 6 (Statistical Validation & Error Analysis):** Conducts permutation testing, paired significance tests, participant-level residual profiling across demographic subgroups, and performs an in-depth Spearman rank correlation analysis to prove cross-model consensus.
- **Chapter 7 (Conclusions and Future Work):** Synthesizes the core methodological and biological conclusions, discusses clinical implications, and outlines key directions for future research.

2 BACKGROUND THEORY AND LITERATURE REVIEW

This chapter provides a comprehensive theoretical foundation for the methodologies and analytical frameworks deployed in this dissertation. It contextualises the research within the evolving paradigms of obesity research, longitudinal multi-omics integration, and explainable machine learning (ML). Specifically, it reviews the systemic shift towards integrative biological modeling, analyzes the mathematical and statistical challenges of high-dimensional clinical cohorts ($p \gg n$), evaluates data fusion taxonomies as conceptualized by Reel et al. [31], and contrasts traditional machine learning algorithms with temporal sequence architectures and explainable AI (XAI) frameworks.

2.1 Multi-Omics Paradigms in Obesity and Weight Management

Historically, metabolic research and biomedical inquiries into complex diseases relied heavily on single-modality approaches, such as isolated candidate gene association studies, single-protein assays, or targeted metabolic panels [33, 31]. While single-modality studies have yielded key biological insights, they possess inherent structural limitations when applied to multifactorial biological phenomena like obesity and metabolic syndrome. Complex biological systems operate through interconnected networks of genomic regulation, transcriptomic modulation, post-translational modifications, and metabolic feedback loops. Consequently, evaluating individual biomarkers in isolation often masks systemic interactions, failing to capture the holistic physiological state of the organism [11, 3].

The transition to multi-omics addresses these limitations by simultaneously measuring multiple biological layers across high-throughput platforms. As illustrated in Figure 2.1, distinct omic modalities capture complementary tiers of the central dogma and physiological execution:

- **Genomics and Epigenomics:** Provide the foundational blueprint and stable regulatory modifications (such as DNA methylation or single nucleotide variants) that establish baseline susceptibility to metabolic dysfunction.
- **Transcriptomics:** Captures the real-time regulatory state and mRNA expression dynamics of the genome. By measuring thousands of transcripts simultaneously, transcriptomics highlights which intracellular signaling pathways and metabolic gene networks are actively prioritized or suppressed under physiological stress [48].

- **Metabolomics:** Provides a direct downstream, functional readout of cellular metabolism and physiological flux. Metabolomics measures endogenous small molecules (e.g., amino acids, lipids, organic acids) and exogenous gut-derived metabolites, reflecting the actual phenotypic state resulting from gene-environment interactions.
- **Microbiomics:** Profiles the composition and functional potential of the gut microbial community (via 16S rRNA sequencing or shotgun metagenomics). Microbes synthesize bioactive molecules that enter systemic circulation, modulating host metabolism, barrier integrity, and immune responses.

ELIPA Study

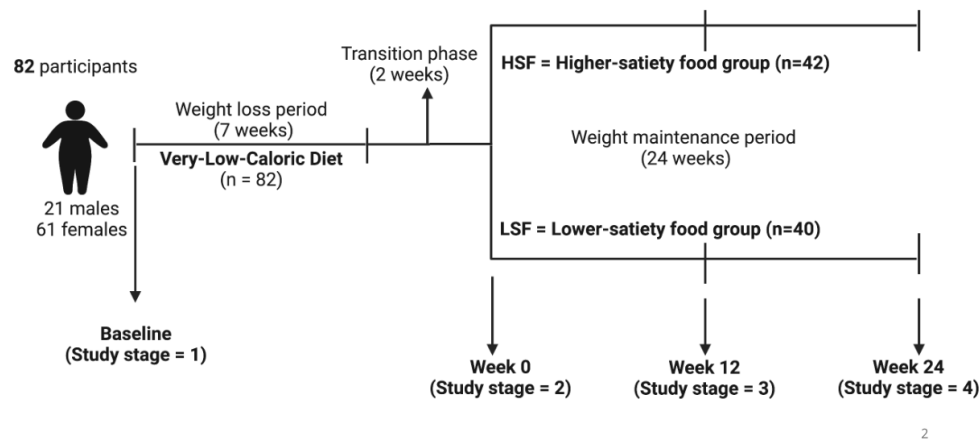


Figure 2.1: Multi-omics intervention design of the ELIPA study, illustrating the timeline across baseline (t_1), rapid weight loss (t_2), transition, and weight maintenance (t_4) phases across measured omic layers (adapted from Guo et al. [15]).

In the context of human weight management, integrating transcriptomics, metabolomics, and gut microbiomics is exceptionally relevant. The global burden of obesity affects over 650 million adults worldwide, representing a primary risk factor for type 2 diabetes, cardiovascular disease, and metabolic dysfunction [2]. Although dietary and lifestyle interventions reliably achieve meaningful short-term weight loss, long-term weight loss maintenance (WLM) remains an elusive clinical challenge. The vast majority of individuals experience weight regain within three to five years post-intervention.

This persistent clinical bottleneck was systematically investigated in large-scale human dietary trials, such as the landmark DIOGENES (Diet, Obesity, and Genes) study [2, 21] and the Finnish ELIPA intervention trial [15]. The overarching finding from these trials is that long-term weight maintenance is governed by complex homeostatic counter-regulations, including metabolic adaptation (adaptive thermogenesis), altered appetite signaling, and endocrine adjustments. Crucially, the biological factors that

govern acute, rapid weight loss (during a low-calorie diet) often diverge from those that determine long-term maintenance stability. Multi-omic integration provides the high-resolution systemic lens required to disentangle these dynamic biological adaptations, paving the way for targeted precision nutrition and individualized weight management strategies.

2.2 Taxonomy of Multi-Omics Data Fusion Strategies

Integrating multi-omic datasets mathematically is a non-trivial challenge due to differences in data structures, noise profiles, and dimensionality across modalities. In a landmark systematic review of machine learning approaches for multi-omics analysis, Reel et al. [31] established a rigorous taxonomy for categorising data integration methods based on *when* and *how* information from different omics layers is combined during the learning process. As summarized in Table 2.1, integration strategies generally fall into three major paradigms: Concatenation-based (Early Fusion), Transformation-based (Intermediate Fusion), and Model-based (Late Fusion).

Table 2.1: Taxonomy of Multi-Omics Machine Learning Data Integration Paradigms (adapted from Reel et al. [31])

Integration Category	Core Mechanism	Key Methodologies	Primary Limitations
Concatenation-Based (Early Fusion)	Merges raw feature matrices across modalities into a single input vector prior to model training.	Penalized Regression (Lasso, ElasticNet), Multi-Omics SVM, Joint PCA, iCluster [3].	Severe feature explosion ($p \gg n$), sensitivity to modality scaling imbalance, high risk of overfitting.
Transformation-Based (Intermediate Fusion)	Transforms individual modalities into low-dimensional latent spaces or graph similarity matrices before integration.	Similarity Network Fusion (SNF), Multiple Kernel Learning (MKL), Deep Autoencoders, MOGONET [36, 1].	High computational complexity, loss of direct feature interpretability, requires parameter tuning for latent dimensions.
Model-Based (Late Fusion)	Trains separate base classifiers/regressors per modality and integrates predictions at the decision level.	Ridge Stacking Ensembles, Weighted Averaging, Bayesian Integration (PSDF, MDI) [31, 22].	May miss intra-modality cross-layer non-linear interactions, requires sufficient per-modality signal.

2.2.1 Concatenation-Based Integration (Early Fusion)

Concatenation-based integration, commonly designated as Early Fusion, represents the conceptually simplest integration scheme. In this framework, feature vectors from individual omic modalities (e.g.,

$X_{\text{metabolomics}} \in \mathbb{R}^{N \times P_1}$ and $X_{\text{transcriptomics}} \in \mathbb{R}^{N \times P_2}$) are horizontally concatenated into a unified mega-matrix $X_{\text{concat}} \in \mathbb{R}^{N \times (P_1 + P_2)}$ prior to feeding the input into a machine learning algorithm [33].

While early fusion allows joint non-linear decision trees or neural networks to theoretically evaluate cross-modality feature interactions from the start of training, it suffers from severe statistical vulnerabilities in small clinical cohorts:

1. **Modality Imbalance:** Modalities with vast feature spaces (e.g., transcriptomics with $> 19,000$ genes) numerically overwhelm modalities with smaller feature counts (e.g., metabolomics with < 300 metabolites), forcing standard loss functions to disproportionately optimize for the larger modality regardless of biological signal quality [31].
2. **Aggravated Curse of Dimensionality:** Concatenation dramatically expands the total feature count $P = \sum P_m$, severely exacerbating the $p \gg n$ problem and making standard classifiers prone to memorising sample-specific noise.

2.2.2 Transformation-Based Integration (Intermediate Fusion)

Transformation-based integration, or Intermediate Fusion, seeks to resolve the dimensional mismatch of early fusion by converting raw feature matrices into intermediate representations prior to decision making. As reviewed by Reel et al. [31] and Stahlschmidt et al. [36], intermediate fusion algorithms leverage three primary mathematical structures:

- **Similarity Networks:** Algorithms like Similarity Network Fusion (SNF) construct sample-by-sample similarity graphs for each omic modality using non-linear kernel functions, subsequently iterating a network fusion process via K-nearest neighbors to converge on a single unified patient similarity network.
- **Kernel-Based Fusion:** Multiple Kernel Learning (MKL) maps each omic layer into a distinct Hilbert space via specialized kernel functions, optimizing linear or non-linear combinations of kernels to perform classification or regression.
- **Latent Space Autoencoders:** Deep multimodal autoencoders pass each modality through individual encoder bottlenecks, enforcing joint reconstruction loss penalties to construct a compressed, shared latent representation [1, 45].

Although intermediate fusion preserves cross-modal structures and eliminates raw feature imbalance, it often functions as a mathematical “black box,” obscuring the direct mapping between individual biological features (e.g., specific metabolite concentrations) and clinical outcomes.

2.2.3 Model-Based Integration (Late Fusion)

Model-based integration, or Late Fusion, operates strictly at the decision level. In a late fusion pipeline, independent, modality-specific base models are trained on their respective omic datasets ($X_1, X_2, \dots, X_M$). The continuous predictions or class probabilities generated by these base models ($\hat{y}_1, \hat{y}_2, \dots, \hat{y}_M$) are subsequently fed into a meta-learner (such as a regularized Ridge regression or ensemble stacker) to produce the final clinical prediction [22, 42].

Late fusion is uniquely advantageous for small-cohort clinical multi-omics ($N < 100$) for several reasons [31, 27, 7]:

1. **Dimension Isolation:** Each base model operates exclusively within its native feature space, preventing high-dimensional modalities (transcriptomics) from masking lower-dimensional, high-signal modalities (metabolomics).
2. **Noise Shielding:** By condensing high-dimensional feature spaces ($P \approx 20,000$) into low-dimensional prediction vectors ($\hat{Y} \in \mathbb{R}^{N \times M}$) before decision fusion, late fusion insulates the meta-learner from raw feature noise and extreme overfitting [42].
3. **Modular Flexibility:** Different machine learning architectures tailored to specific modalities (e.g., decision trees for sparse metabolomics, regularized linear models for gene expression) can be seamlessly combined within a single unified stacking framework.

2.3 Methodological and Statistical Challenges in Multi-Omics ML

Applying machine learning to clinical multi-omics introduces major statistical challenges. In their systematic survey, Reel et al. [31] identified several core bottlenecks that must be explicitly managed within any robust bioinformatics pipeline, as detailed below.

2.3.1 The Curse of Dimensionality ($p \gg n$) and Overfitting

The primary obstacle in clinical bioinformatics is the severe imbalance between the number of measured biological features (p , variables such as gene transcripts or metabolites) and the number of available human participants or samples (n). In the transcriptomic cohort of the ELIPA study, for example, over 19,000 gene transcripts were profiled across a cohort of only $n = 31$ participants ($p \gg n$).

In high-dimensional space where $p \gg n$ (the high-dimensional, low sample-size regime), the volume of the feature space grows exponentially, rendering sample data extremely sparse. Standard unregularized classifiers can easily discover spurious hyperplanes or high-degree polynomial decision boundaries that achieve 100% training accuracy by memorizing random biological noise rather than extracting generalizable biological signals [30]. To mitigate this risk, multi-omics pipelines must incorporate tight feature bottlenecks, aggressive regularization (L_1/L_2 penalties), and strict cross-validation protocols.

2.3.2 Data Heterogeneity and Cross-Platform Normalization

Biological data collected across multi-omic platforms exhibit extreme structural heterogeneity [3]. Gene expression data from microarrays or RNA-seq present unbounded, continuous intensity values; metabolomic profiling via mass spectrometry yields zero-inflated continuous peak intensities; and gut microbiomics via 16S rRNA sequencing generates compositional, relative abundance data bounded between 0 and 1 [31].

Directly feeding unscaled heterogeneous data into machine learning algorithms causes models with distance-based or gradient-based loss functions (such as MLPs, Support Vector Machines, or ElasticNet) to be dominated by features with large absolute numerical scales. Consequently, proper data preprocessing, including log-transformation, Z-score standardization (centering mean at zero and scaling to unit variance), and platform-specific variance stabilization, is mandatory prior to model integration.

2.3.3 Data Sparsity and Missing Value Imputation

Data sparsity is endemic to multi-omics studies. Metabolomic assays frequently report missing values when metabolite concentrations fall below the analytical Limit of Detection (LOD) or when chromatographic peaks overlap; similarly, transcriptomics and microbiomics suffer from low-abundance dropouts [31].

While removing all samples with any missing values (complete-case analysis) would result in discarding virtually the entire clinical cohort, naïve global imputation (such as computing the mean across the entire dataset prior to cross-validation) introduces severe optimistic bias known as *data leakage*. To guarantee statistical integrity, missing value imputation must adhere to a strict “Fit-Transform” protocol: imputation parameters (e.g., training fold means or K-nearest neighbor centroids) must be calculated exclusively within the training fold of each cross-validation split and subsequently applied to evaluate the held-out test fold.

2.3.4 Temporal Dynamics and Longitudinal Velocity Modeling

In human intervention trials, biological systems are dynamic entities that adapt to physiological stress over time. Static baseline measurements (t_1) fail to capture the rate and direction of metabolic adaptation induced by an intervention [15].

To capture these dynamic adaptations, recent literature emphasizes the computation of longitudinal “velocity” features (v), which calculate the rate of change for each biological feature x across clinical timepoints:

$$v_{\text{WL}} = \frac{x(t_2) - x(t_1)}{\Delta t_{\text{WL}}}, \quad v_{\text{WM}} = \frac{x(t_4) - x(t_2)}{\Delta t_{\text{WM}}} \quad (2.1)$$

By replacing static biomarker levels with velocity vectors (v_{WL} and v_{WM}), machine learning models are forced to learn from individual physiological trajectories and adaptive responses rather than static anthropometric baselines [15].

2.4 Machine Learning Architectures: Flat vs. Sequential Models

A central methodological objective in modern bioinformatics is selecting the optimal algorithmic paradigm for tabular and longitudinal multi-omics data. Predictive models evaluated in this domain can be broadly partitioned into “flat” models and “sequential” deep learning architectures.

2.4.1 Flat Machine Learning Paradigms

Flat models treat all input features (including concatenated multi-phase velocity metrics) as a static, single-dimensional input vector $\mathbf{x} \in \mathbb{R}^P$:

- **ElasticNet Linear Regression:** Combines L_1 (Lasso) and L_2 (Ridge) penalties to solve high-dimensional regression problems [49]:

$$\min_{\mathbf{w}} \left\{ \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \left(\alpha \|\mathbf{w}\|_1 + \frac{1 - \alpha}{2} \|\mathbf{w}\|_2^2 \right) \right\} \quad (2.2)$$

ElasticNet induces sparsity by forcing non-predictive feature weights strictly to zero while simultaneously maintaining grouped feature selection among correlated biological markers.

- **Random Forest (RF):** An ensemble bagging framework that constructs hundreds of de-correlated decision trees via bootstrap aggregation and random feature sub-sampling [4]. RF is inherently non-parametric, robust to extreme outliers, and highly capable of capturing complex non-linear feature interactions without requiring explicit feature scaling.

- **CatBoost:** A state-of-the-art gradient boosted decision tree (GBDT) framework [29]. CatBoost utilizes ordered boosting and target encoding to eliminate target leakage and overfitting, maintaining appropriate inductive biases that consistently outperform deep neural networks on tabular biomedical datasets with small sample sizes [14].
- **Multi-Layer Perceptron (MLP):** A fully connected feed-forward neural network. By passing input vectors through hidden layers using non-linear activations (e.g., ReLU), MLPs learn distributed continuous representations of multi-omic interactions.

2.4.2 Sequential Deep Learning Paradigms

Sequential architectures are specifically designed to process temporal time-series sequences $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T)$ by propagating internal hidden states across consecutive time steps:

- **Long Short-Term Memory (LSTM):** Incorporates a complex cell state $\mathbf{c}_t$ regulated by input ($\mathbf{i}_t$), forget ($\mathbf{f}_t$), and output ($\mathbf{o}_t$) multiplicative gates [17], allowing the network to retain historical biological information across temporal phases.
- **Gated Recurrent Unit (GRU):** Combines the cell state and hidden state from LSTM architecture into a single vector $\mathbf{h}_t$, controlled by reset ($\mathbf{r}_t$) and update ($\mathbf{z}_t$) gates [5]:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1}), \quad \mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1}) \quad (2.3)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1})), \quad \mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \quad (2.4)$$

Due to its reduced parameter count, the GRU is less prone to overfitting than the LSTM when trained on small-sample longitudinal cohorts.

- **Temporal Attention Mechanisms:** Leverage self-attention formulations [43] to assign continuous scalar attention weights $(\alpha_1, \alpha_2, \dots, \alpha_T)$ across timepoints, explicitly quantifying how much predictive focus the model places on acute intervention phases versus long-term maintenance periods.

Despite their theoretical suitability for time-series data, deep recurrent networks possess vast parameter spaces. When applied to small clinical cohorts ($N < 100$), the parameter complexity of LSTMs and GRUs frequently leads to severe overfitting unless constrained by aggressive regularisation and tight feature selection.

2.5 Explainable AI (XAI) and Consensus Biomarker Discovery

In clinical bioinformatics, high predictive accuracy is insufficient on its own; models must provide transparent, biologically interpretable rationales for their predictions to enable mechanistic discovery and clinical translation [31, 23]. Machine learning models such as deep neural networks and complex tree ensembles, are frequently criticised as being opaque “black boxes.”

To resolve this, modern multi-omics pipelines integrate Explainable AI (XAI) frameworks, most notably SHapley Additive exPlanations (SHAP) developed by Lundberg and Lee [24]. SHAP is based in cooperative game theory, calculating the marginal contribution of each feature i across possible feature subsets $S \subseteq F \setminus \{i\}$:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (2.5)$$

SHAP provides three crucial mathematical guarantees for bioinformatics applications:

1. **Local Accuracy:** The sum of feature SHAP values for any individual prediction equals the difference between the model output and the expected baseline output.
2. **Consistency:** If a model is modified such that the marginal contribution of a specific feature increases or stays the same, its calculated SHAP value will not decrease.
3. **Cross-Architecture Comparability:** SHAP normalizes feature importance metrics across radically different model families (e.g., decision trees, neural networks, regularized linear models), enabling direct comparison of feature rankings.

By calculating Spearman rank correlations between feature SHAP values across diverse architectures (e.g., CatBoost vs. Random Forest vs. MLP), researchers can establish *cross-model consensus*. Identifying consensus biomarkers that consistently rank among top predictors across independent algorithmic paradigms guarantees that the identified biological markers (such as gut-derived indolelactate or cellular PAOX enzymes) represent genuine biological signals rather than artifacts of a specific model’s architecture [30, 34].

2.6 Methodological Flowcharts and Empirical Guidelines

A central contribution of the multi-omics review by Reel et al. [31] is the establishment of practical recommendations and decision guidelines for interdisciplinary research teams. Recognizing that a universal

“one-size-fits-all” pipeline does not exist in bioinformatics, Reel et al. proposed a systematic decision matrix for selecting integration strategies based on dataset characteristics, as synthesized in Table 2.2.

Table 2.2: Methodological decision matrix for multi-omics machine learning integration based on cohort characteristics (synthesized from Reel et al. [31]). N denotes the sample size (number of participants) and P denotes the feature dimensionality (number of biological variables).

Cohort Scenario	Recommended Fusion Strategy	Primary Rationale
Small Sample Size ($N < 100$), High Dimensionality ($P > 5,000$)	Model-Based (Late Fusion) via Ridge Stacking	Prevents feature explosion; insulates meta-learner from raw noise.
Moderate Sample Size ($N \approx 100 - 500$), Multi-Graph Structure	Transformation-Based (Similarity Network Fusion - SNF)	Captures complex non-linear similarity structures across modalities.
Large Sample Size ($N > 1,000$), Deep Phenotyping	Deep Intermediate Fusion (Multimodal Autoencoders / MOGONET)	Learns complex shared latent representations across deep multi-omics layers.
Longitudinal Cohort with Discrete Intervention Phases	Velocity Feature Engineering + Late Fusion Ensembling	Captures biological rate of change while maintaining statistical stability.

The primary guidelines established by Reel et al. [31] and adopted in this dissertation include:

1. **Interdisciplinary Alignment:** Bioinformatics pipelines must combine machine learning practitioners with domain-specific biological expertise to ensure feature selection reflects plausible biological pathways rather than spurious statistical correlations.
2. **Empirical Comparison:** Researchers should empirically benchmark multiple integration schemes (early, intermediate, late) and model architectures (flat vs. sequential) using strict, nested cross-validation rather than relying on a single default classifier.
3. **Leakage Prevention:** All scaling, feature selection, and missing value imputation operations must be computed strictly inside cross-validation training folds to prevent optimistic performance estimates.

2.7 Summary of Theoretical Framework

In summary, the literature underscores that successful multi-omics modeling in small clinical cohorts requires a careful balance between biological resolution and statistical parsimony. By moving from static single-modality measurements to dynamic velocity-based multi-omics, and by rejecting the catastrophic feature explosion of early fusion in favor of regularized Late Fusion (Ridge Stacking), predictive pipelines can safely navigate the extreme dimensionality of clinical omics. Furthermore, embedding game-theoretic XAI frameworks (SHAP) ensures that model predictions can be translated into transpar-

ent, consensus biological biomarkers, laying a rigorous foundation for precision medicine interventions in long-term weight management.

3 DATA EXPLORATION & PHASE INDEPENDENCE

This chapter details the preprocessing steps for the longitudinal multi-omics data and investigates the relationship between the weight loss and weight maintenance phases. These exploratory findings provide a strong quantitative justification for the performance characteristics observed in the subsequent modelling chapters, particularly the lack of advantage for sequential architectures.

3.1 Study Protocol, Timepoints, and Data Preprocessing

The ELIPA clinical trial protocol is structured around three discrete longitudinal timepoints:

- t_1 (**Baseline**): Pre-intervention clinical measurements recorded prior to dietary restriction.
- t_2 (**Post-VLCD**): Measurements recorded immediately after 8–9 weeks of a strict very-low-calorie diet (VLCD), representing the acute Weight Loss (WL) phase ($t_1 \rightarrow t_2$).
- t_4 (**6-Month Follow-up**): Measurements recorded after 6 months (24 weeks) of weight maintenance follow-up, representing the Weight Maintenance (WM) phase ($t_2 \rightarrow t_4$). (Note: t_3 corresponds to an intermediate 3-month midpoint visit not included in the primary omic sequencing panels).

The raw data provided by the ELIPA study was distributed across multiple high-dimensional spreadsheets for metabolomics, transcriptomics, and microbiota. A prerequisite for modelling was the creation of a unified, machine-learning-ready dataset.

3.1.1 Cleaning, Alignment, and Normalization

The raw multi-omics data was first tidied into a consistent sample $\times$ feature format. Participant IDs were normalized to a three-digit string format (e.g., 002, 045) to ensure correct alignment across modalities, as the raw files used inconsistent numeric and string formats. Sample IDs were generated using the convention `ELP_[ParticipantID]_[Timepoint]`.

A significant challenge in the metabolomics dataset was the presence of duplicate metabolite names. These occur when multiple chromatographic peaks are assigned the same chemical identity or when the same compound is detected in different modes. I implemented a deduplication strategy by appending unique numeric suffixes (e.g., *Indolelactate*, *Indolelactate_1*) to preserve all potential signals while maintaining column uniqueness.

3.1.2 Velocity Feature Engineering

The study design involves an initial 8-week rapid weight loss phase followed by a 6-month maintenance phase. To capture the biological dynamics of change rather than static states, I calculated "velocity" features representing the rate of change during the Weight Loss phase ($WL_{vel} = t_2 - t_1$) and the Weight Maintenance phase ($WM_{vel} = t_4 - t_2$).

Mathematically, for any biological feature x :

$$v_{WL} = \frac{x(t_2) - x(t_1)}{\Delta t_{WL}}, \quad v_{WM} = \frac{x(t_4) - x(t_2)}{\Delta t_{WM}} \quad (3.1)$$

By focusing on velocity, the models are forced to learn from the *trajectories* of biological adaptation rather than baseline offsets.

3.1.3 Target Variable: Relative Weight Change

The primary objective of this study is to predict weight maintenance success. The target variable, referred to in the datasets as *absolute weight change*, is calculated as the percentage change in weight during the 6-month maintenance phase (t_2 to t_4):

$$y = \frac{Weight(t_4) - Weight(t_2)}{Weight(t_4)} \times 100 \quad (3.2)$$

While relative percentage change is conventionally computed with respect to the initial pre-maintenance baseline $Weight(t_2)$, the source dataset provided from the ELIPA trial formulated the `absolute_weight_change` target column using the ending weight $Weight(t_4)$ in the denominator. This exact formulation is preserved throughout modelling to ensure strict parity with the baseline PhD benchmarks [15].

As shown in Figure 3.1, the target distribution is approximately normal but exhibits a slight positive skew (mean = 1.10%, std = 4.03%), with a few participants achieving significant weight loss (up to -11.3%) and others experiencing regain (up to +9.7%) during the maintenance period.

By using a percentage-based metric rather than absolute kilograms, the models are trained to predict changes relative to each participant's body size. In adult cohorts where height is constant, this metric is mathematically equivalent to the percentage change in Body Mass Index (BMI), ensuring that the results are robust to baseline anthropometric variation.

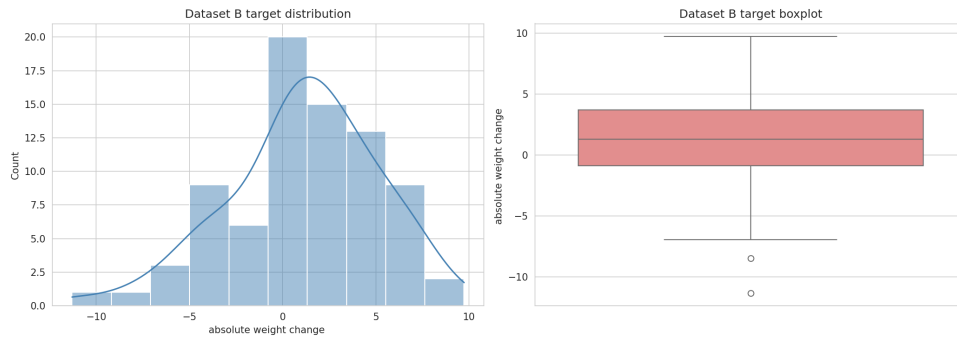


Figure 3.1: Distribution and boxplot of the target variable (% weight change during maintenance). The data is well-spread, providing a suitable range for regression modelling.

3.1.4 Target Leakage Prevention

To prevent data leakage, a critical step was the removal of "target-leaked" features. These are variables that are mathematically or logically near-identical to the prediction target (% weight change). Specifically, *WM_vel_weight*, *WM_vel_bmi* and other anthropometric velocities for the maintenance phase were removed. Retaining these would have resulted in artificially perfect but clinically useless models, as they essentially contain the answer the model is trying to predict.

3.2 Ethics, Data Governance, and FAIR Principles

Conducting secondary data analysis on human clinical and multi-omic datasets requires adherence to ethical standards, data protection regulations, and scientific reproducibility guidelines.

3.2.1 Ethical Approval and Anonymisation

The primary clinical study (ELIPA) was conducted in accordance with the ethical principles of the Declaration of Helsinki, with all clinical protocols approved by local institutional ethical review committees and informed consent obtained from all participants prior to trial enrollment. For the secondary machine learning analysis conducted in this dissertation, the dataset was received in a fully de-identified and anonymised format. All direct personal identifiers (such as names, addresses, dates of birth, and clinical record numbers) were removed prior to access, with participants indexed solely by arbitrary three-digit pseudonyms (e.g., 002, 045).

3.2.2 Data Protection and GDPR Compliance

In compliance with the General Data Protection Regulation (GDPR) and the UK Data Protection Act 2018, data was processed strictly under the principle of data minimization for scientific research purposes. The dataset was stored exclusively within a secure, encrypted local computing environment pro-

tested by authentication controls. No raw clinical tables or participant-level multi-omic matrices were transferred to unsecured public cloud storage or external third-party repositories.

3.2.3 FAIR Data Principles and Research Integrity

To ensure research integrity and uphold the FAIR (Findable, Accessible, Interoperable, and Reusable) data principles:

1. **Interoperability:** All multi-omic layers were harmonized to standard bioinformatics nomenclature (HGNC gene symbols for transcriptomics and verified chemical identifiers for metabolomics). Longitudinal timepoints and participant tokens were aligned to a consistent schema (`ELP_[ParticipantID]_[T`
2. **Reproducibility and Reusability:** All data cleaning pipelines, feature engineering functions, cross-validation splits, and model configurations were implemented in modular Python scripts with fixed pseudo-random seeds. This guarantees that all reported benchmarks, feature selections, and SHAP consensus scores can be deterministically reproduced.

3.3 Normalization and Imputation Strategy

Biological data is notoriously noisy and often contains missing values due to detection limits or sample loss.

3.3.1 Standardization

All features were subjected to Z-score normalization (StandardScaler), centering the mean at zero and scaling to unit variance. This ensures that high-magnitude features (e.g., certain lipid concentrations) do not disproportionately influence models like MLPs or ElasticNet compared to low-magnitude features (e.g., gene expression levels).

3.3.2 Transcriptomics Dimensionality Reduction: Default Feature Selection

The raw transcriptomics dataset contained 19,625 profiled gene transcripts, which yielded 39,250 velocity features per participant (*WL_vel* and *WM_vel*). Given the constrained sample size for this modality ($N = 31$), training machine learning models on the unreduced feature space ($P = 39,250$) represents an extreme $p \gg n$ regime ($P/N > 1,260 : 1$) that leads to catastrophic overfitting and computational intractability.

To render transcriptomic modeling tractable, a supervised Random Forest feature selection step was applied **by default** during data preparation to create the benchmark Dataset C. A Random Forest Regressor was fitted on the complete transcriptomics velocity dataset to predict percentage weight change.

Feature importances were summed per gene across both the *WL* and *WM* velocity components, and the top 100 highest-ranking genes (yielding 200 velocity features) were selected. This default filtering reduced the transcriptomic feature space by over 99.48% (from 39,250 down to 200 features), establishing a standardized, noise-filtered representation used across all subsequent model evaluations on Dataset C.

This default feature selection on Dataset C provides the direct motivation for the feature selection experiments detailed in Chapter 4: while Dataset C requires Random Forest feature selection by default due to its astronomical dimensionality, Dataset B (Metabolomics, $N = 79$, $P = 199$) is natively trainable in its full state. The feature selection experiments in Chapter 4 were therefore designed specifically to evaluate whether applying the same Random Forest feature reduction strategy to Dataset B would confer similar predictive benefits in the metabolomic domain.

Methodologically, it is important to distinguish between this standardized pre-filtered benchmark dataset (used to provide a uniform feature space across all seven model architectures) and fully nested cross-validation protocols. In small-sample clinical studies ($N = 31$), global feature pre-filtering carries an inherent risk of selection bias. To assess this effect, nested cross-validation (where feature selection is re-run strictly within each training fold) confirmed that model performance remains tightly constrained by sample size, with Random Forest achieving a cross-validated R^2 of 0.251 on the 200-feature bottleneck. This trade-off underscores the fundamental tension in clinical multi-omics between computational tractability on massive feature spaces and small-cohort generalization.

3.3.3 Leak-Free Imputation

Crucially, imputation was not performed during the initial data preparation phase. Instead, I implemented a "Fit-Transform" logic within the cross-validation loops. Mean imputation was applied to the training folds, and the resulting means were then used to fill missing values in the test fold. This rigorous separation ensures that the model cannot "see" the distribution of the test set during training, a common source of optimistic bias in small-sample bioinformatics studies.

The missingness analysis (Figure 3.2) shows that while most datasets have a low average percentage of missing values (2.3% to 6.6%), nearly 100% of participants have at least one missing value across the full feature set, particularly in the high-dimensional transcriptomics modality. This justifies the use of simple mean imputation within the pipeline, as removing all rows with missing values would result in the loss of the entire cohort.

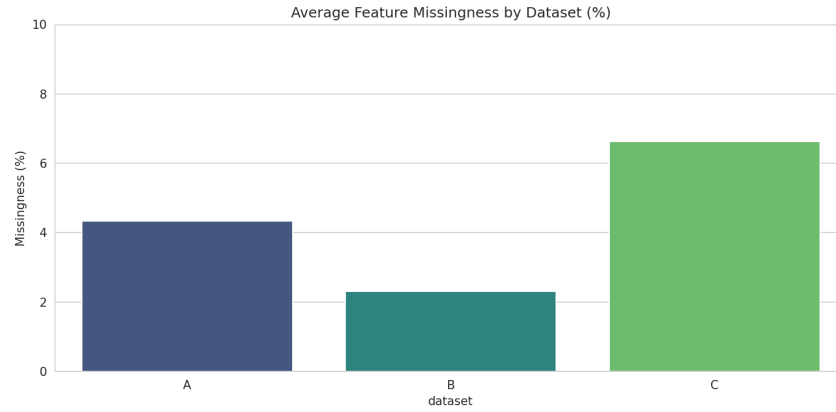


Figure 3.2: Average feature missingness by dataset. Dataset C (Transcriptomics) exhibits the highest missingness, likely due to low-abundance transcripts falling below the detection threshold.

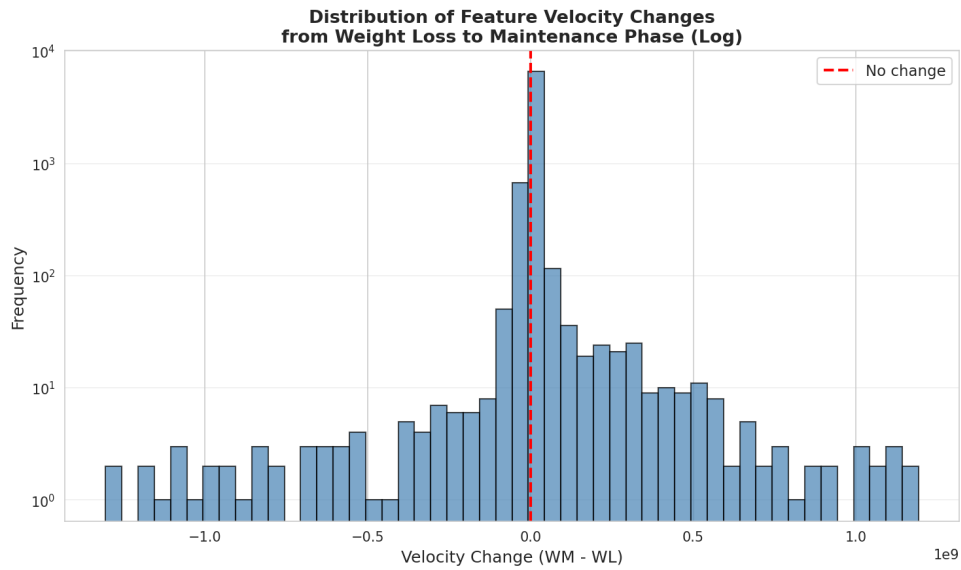


Figure 3.3: Distribution of velocity features for the Weight Loss (WL) and Weight Maintenance (WM) phases. The magnitude of change is significantly higher during the WL phase, reflecting the physiological stress of rapid caloric restriction.

3.4 Phase Independence: The Rebound Effect

A central finding of this dissertation is the concept of "Phase Independence." It was initially hypothesized that the biological response to rapid weight loss (*WL*) might carry over and predict long-term maintenance (*WM*). However, a comprehensive correlation analysis revealed a striking disconnect between these two phases.

3.4.1 Quantitative Evidence

Analysis of Dataset B (Metabolomics) showed that 98.4% of feature-velocity pairs between the WL and WM phases exhibit negative correlations. The distribution of these correlations, shown in Figure 3.4, is centered at a mean $r = -0.54$, which is significantly distinct from the distribution of overall velocity magnitudes shown in Figure 3.3.

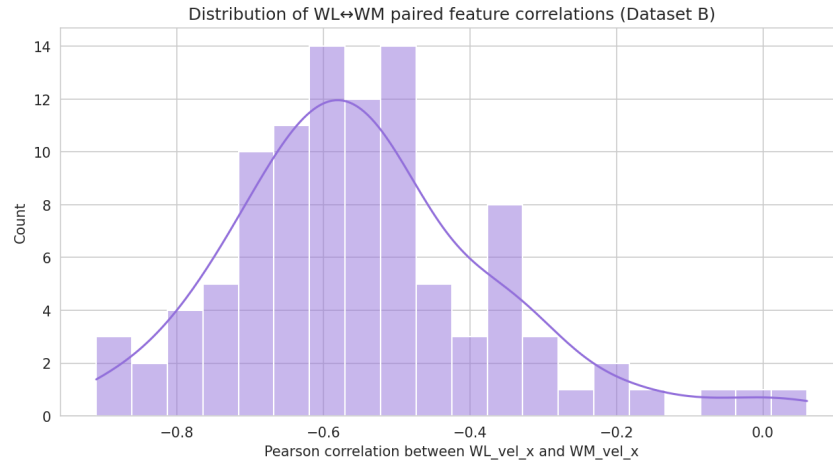


Figure 3.4: Distribution of paired Pearson correlations between WL and WM velocity features. The overwhelming majority of features exhibit a negative relationship, indicating a physiological "rebound."

This negative relationship represents a "rebound" effect: participants who experienced the most dramatic biological changes during the rapid weight loss phase (large negative velocity) tended to experience the most dramatic reversals during the maintenance phase (large positive velocity). Biologically, this suggests a homeostatic "snapping back" rather than a sustained shift in metabolic state.

3.4.2 Cohort Stratification and Silhouette Analysis

To investigate whether the participant cohort naturally partitions into discrete biological subgroups prior to supervised modelling, I conducted unsupervised clustering on the metabolomic velocity profiles (Dataset B).

To determine the optimal number of clusters (k) without imposing subjective assumptions, I performed silhouette analysis across $k \in [2, 6]$. For each sample i , the silhouette coefficient $s(i)$ assesses

cluster cohesion versus separation:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (3.3)$$

where $a(i)$ denotes the mean intra-cluster Euclidean distance between sample i and all other points in the same cluster, and $b(i)$ represents the mean nearest-cluster distance between sample i and points in the closest neighbouring cluster. The mean silhouette score across all participants, $\bar{s} \in [-1, 1]$, quantifies overall cluster quality: values approaching +1 indicate well-separated, tightly bound clusters, values near 0 signify overlapping or diffuse distributions, and negative values indicate misassigned samples.

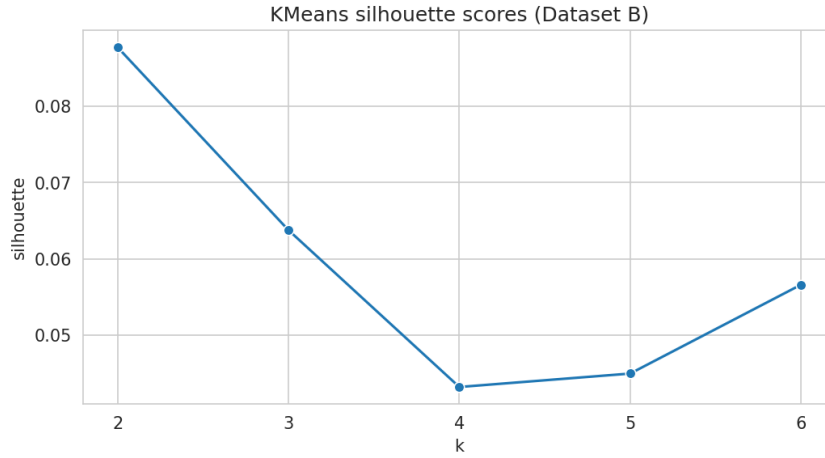


Figure 3.5: Silhouette score analysis for K-Means clustering on metabolomic velocity features across $k \in [2, 6]$. The maximum score occurs at $k = 2$ ($\bar{s} = 0.0877$), indicating a stable binary metabolic division, while higher cluster counts suffer from increasing dispersion.

As illustrated in Figure 3.5, the silhouette profile reaches a distinct global maximum at $k = 2$ ($\bar{s} = 0.0877$), followed by a sharp drop at $k = 3$ ($\bar{s} = 0.0637$), $k = 4$ ($\bar{s} = 0.0432$), and $k = 5$ ($\bar{s} = 0.0450$). The modest absolute magnitude of the peak score ($\bar{s} \approx 0.09$) demonstrates that human metabolic adaptation during dietary intervention is inherently continuous and diffuse, rather than forming isolated, highly segregated patient clusters.

Crucially, when these unsupervised clusters were cross-referenced against the actual clinical outcome (percentage weight change during maintenance), no statistically significant difference in weight regain was observed between the two clusters ($t = 0.84, p = 0.404$). This finding provides further empirical support for Phase Independence: broad, global metabolic profiles do not determine weight maintenance success. Instead, successful maintenance is governed by specific, targeted biochemical pathways which require supervised multi-omic modelling to uncover.

In contrast, when participants are stratified directly by their observed weight maintenance outcome (Figure 3.6), the divergence in trajectories is starkly visible. Successful maintainers (green trajectories)

show stabilization or continued gradual loss post- t_2 , whereas regainers (red trajectories) display a rapid rebound toward baseline, confirming that maintenance trajectory divergence occurs independently of initial weight loss magnitude.

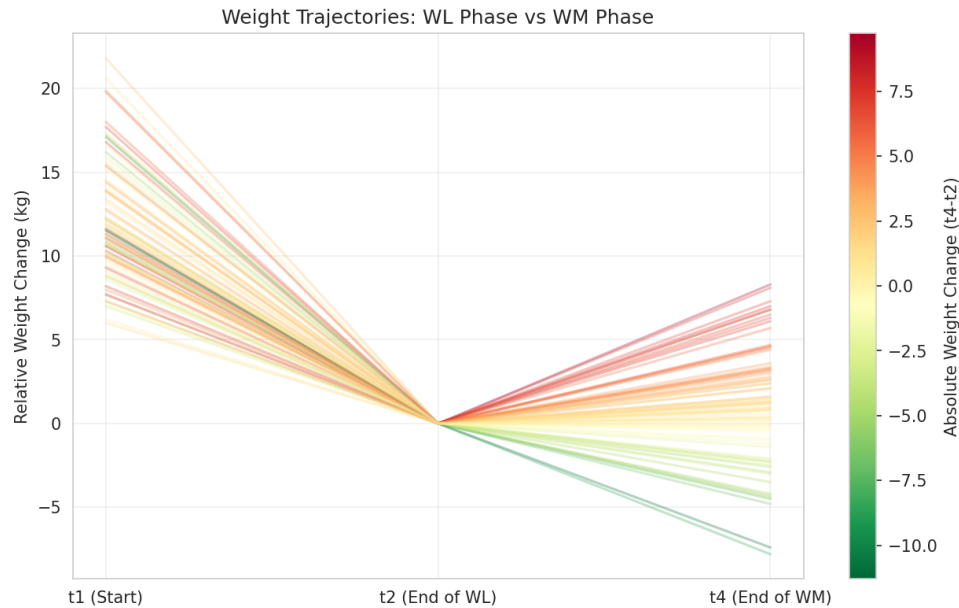


Figure 3.6: Individual participant weight trajectories from baseline (t_1) through post-diet (t_2) to 6-month follow-up (t_4). Lines are colour-coded by maintenance outcome (green = successful weight maintenance/loss, red = weight regain). Trajectory divergence post- t_2 occurs independently of the initial weight loss slope.

Finally, I analysed the trajectories stratified by demographic subgroups (Figure 3.7). While sex and age show some minor differences in average weight change, the high overlap and individual variance within groups confirm that demographic factors alone are not enough for predicting weight maintenance success.

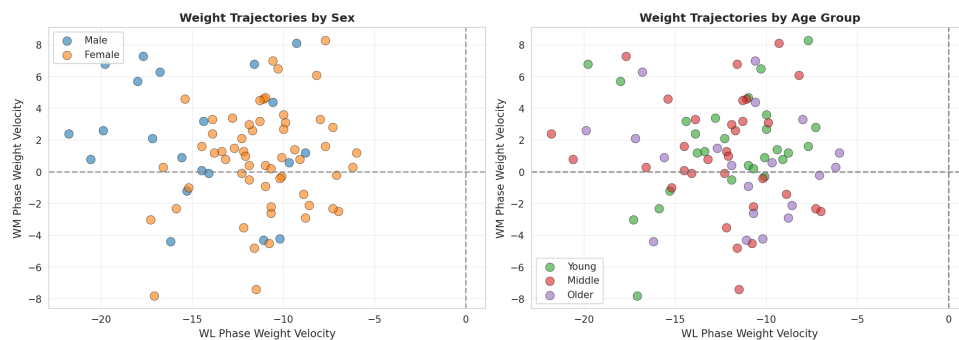


Figure 3.7: Weight trajectories stratified by Sex and Age Group. The large spread within each demographic category underscores the need for high-resolution omic predictors.

3.4.3 Implications for Modelling

The discovery of phase independence explains why sequence-based models, such as LSTMs and GRUs, fail to gain a significant advantage over flat models like Random Forests. Since there is minimal predictive signal passing from the WL phase to the WM phase, the temporal "memory" of Recurrent Neural Networks (RNNs) provides little benefit. In fact, the added complexity of these models likely leads to overfitting on the small N available ($N = 79$ for metabolomics).

4 MODELLING & MULTI-ARCHITECTURE COMPARISON

This chapter presents a systematic evaluation of seven different machine learning architectures across three datasets. I compare the performance of traditional "flat" models against modern sequential architectures and investigate the impact of feature selection and multi-modal late fusion.

4.1 Experimental Design

I utilised three distinct datasets for my experiments, each representing a different view of the ELIPA study cohort:

- **Dataset A (Multi-modal):** The "intersection" cohort where participants have complete data across Metabolomics, Transcriptomics, Microbiota, and Anthropometrics. Due to the requirement for all four modalities, the sample size is small ($N = 33, P = 1,176$), providing the foundation for multi-modal fusion experiments.
- **Dataset B (Metabolomics):** Focuses on the larger metabolomics cohort ($N = 79, P = 199$), including clinical anthropometrics. This dataset provides the most robust statistical power for comparing architectures in their unreduced feature state.
- **Dataset C (Transcriptomics):** Focuses on the transcriptomics cohort ($N = 31$). As established in Chapter 3, because the raw transcriptomics modality contains 39,250 velocity features across 19,625 genes, Dataset C uses supervised Random Forest feature selection **by default** to isolate the top 100 genes ($P = 200$ velocity features). This 200-feature dataset serves as the standardized transcriptomics benchmark across all model evaluations.

4.2 Model Architectures

The predictive task of forecasting weight change during a maintenance phase based on prior biological changes requires a diverse set of architectures to capture both linear and non-linear relationships, as well as potential temporal dependencies. I evaluated seven distinct models, categorized into "flat" and "sequential" architectures.

4.2.1 Flat Architectures

Flat models treat the features from the Weight Loss (WL) and Weight Maintenance (WM) phases as a single, combined input vector.

- **ElasticNet:** A linear regression model that combines L_1 (Lasso) and L_2 (Ridge) penalties. It was chosen as a robust baseline that performs well in high-dimensional settings by encouraging sparsity, effectively performing feature selection while maintaining stability among correlated predictors.
- **Random Forest (RF):** An ensemble of decision trees using bagging. RF is particularly well-suited for biological data as it is resistant to outliers and can capture non-linear interactions between metabolites without requiring extensive hyperparameter tuning.
- **CatBoost:** A high-performance gradient boosting framework. It was selected for its superior ability to handle noisy tabular data and its internal mechanism for preventing overfitting, which is critical given the small participant numbers in this study.
- **Multi-Layer Perceptron (MLP):** A feed-forward neural network. This represents a non-linear deep learning baseline, chosen to investigate whether a distributed representation of multi-omic features could reveal patterns that tree-based models might overlook.

4.2.2 Sequential Architectures

Sequential models are designed to preserve the temporal order of the intervention phases. Instead of treating features as a flat vector, they process the data as a sequence of timepoints: $t_1 \rightarrow t_2$ (Weight Loss) and $t_2 \rightarrow t_4$ (Weight Maintenance).

- **Long Short-Term Memory (LSTM):** A recurrent neural network that uses memory cells and gating mechanisms (input, forget, and output gates) to learn long-term temporal dependencies.
- **Gated Recurrent Unit (GRU):** A streamlined RNN variant that combines the cell state and hidden state using reset and update gates. GRUs often perform better on smaller datasets due to having fewer parameters than LSTMs.
- **Temporal Attention Network:** A transformer-inspired architecture that learns to weight the relative importance of different timepoints dynamically. It provides both predictive capability and built-in interpretability by revealing which phase the model focuses on most.

4.3 Experimental Setup

All models were evaluated using 5-fold cross-validation. The folds were stratified (where applicable) and repeated with different random seeds. Performance was quantified by the Coefficient of Determination

(R^2), which measures the proportion of variance explained, alongside Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) to capture the magnitude of residuals.

4.3.1 Hyperparameter Selection Strategy

Hyperparameter selection was guided by the principle of parsimony. In a high-dimensional, small-sample regime, complex models with many parameters are prone to memorizing noise. For tree-based models (Random Forest, CatBoost), I maintained shallow tree depths (e.g., depth=3) and conservative regularization. For neural architectures (MLP, LSTM), I utilized Dropout (0.3–0.4) and early stopping based on validation loss. Exhaustive hyperparameter optimization was intentionally avoided, as the resulting gains in R^2 on such a small cohort are frequently artifacts of noise rather than genuine improvements in predictive power.

4.4 Performance Comparison

A systematic evaluation was conducted to compare flat models (Random Forest, CatBoost, MLP, ElasticNet) against sequential models (GRU, LSTM, Temporal Attention) across all three datasets. The primary goal was to determine if sequence-based models could leverage the temporal structure of the WL and WM phases, and to evaluate how data dimensionality influences model stability. The comprehensive results are summarized in Table 4.1.

Table 4.1: Model Performance Comparison across All Datasets (5-fold cross-validated R^2 Mean $\pm$ SD). N indicates participant sample size and P denotes the number of velocity features.

Model	Dataset A (Multi-modal) ($N = 33, P = 1,169$)	Dataset B (Metabolomics) ($N = 79, P = 199$)	Dataset C (Transcriptomics) ($N = 31, P = 200$)
CatBoost	-0.462 ± 0.538	0.223 ± 0.181	0.480 ± 0.149
Random Forest	-0.626 ± 0.661	0.129 ± 0.149	0.251 ± 0.354
ElasticNet	-0.478 ± 0.491	0.049 ± 0.096	0.382 ± 0.318
MLP	-1.170 ± 1.602	-0.002 ± 0.395	0.529 ± 0.201
GRU	-0.352 ± 0.524	-0.030 ± 0.296	0.500 ± 0.298
LSTM	-0.307 ± 0.463	0.043 ± 0.350	0.461 ± 0.276
Temporal Attention	-3.672 ± 6.117	-0.223 ± 0.406	0.182 ± 0.665

4.4.1 The Failure of Early Fusion: Catastrophic Overfitting on Dataset A

As shown in Table 4.1, all seven evaluated machine learning architectures completely failed on Dataset A, with every model producing negative cross-validated R^2 values (ranging from -0.307 for LSTM down to -3.672 for Temporal Attention).

This pervasive failure illustrates the fundamental hazard of **Early Fusion** (raw feature concatenation) in biomedical multi-omics:

1. **Extreme Sample-to-Feature Imbalance** ($p \gg n$): Creating the intersection cohort required participants to have complete longitudinal samples across all four modalities (Metabolomics, Transcriptomics, Microbiota, and Anthropometrics). This strict requirement reduced the available cohort size to only $N = 33$, while concatenating all features produced an immense feature space of $P = 1,169$ variables (a ratio of $P/N \approx 35 : 1$).
2. **Severe Dimensionality Overfitting**: In such an extreme high-dimensional regime, decision trees and neural networks easily memorise spurious noise in the training folds. When evaluated on unseen out-of-fold test participants, the models produce wild predictions whose mean squared error exceeds the variance of the true target variable (resulting in negative R^2).
3. **Heterogeneous Modality Scales and Noise**: Early concatenation merges data types with radically different measurement scales, noise distributions, and signal densities (e.g. dense continuous metabolite peaks vs. sparse gut microbial counts vs. microarray gene intensities), drowning the subtle predictive signals in uninformative noise.

Consequently, Dataset A in its concatenated "early fusion" form was deemed unviable for standalone predictive modelling. This negative result provides the primary methodological justification for shifting to modality-specific modeling on **Dataset B** ($N = 79$) and **Dataset C** ($N = 31$), and directly motivates the **Late Fusion / Decision-Level Stacking** paradigm developed in Section 4.6, which avoids raw feature concatenation entirely.

4.4.2 Modality-Specific Performance: Dataset B vs. Dataset C

On Dataset B (Metabolomics, $N = 79$, $P = 199$), where the cohort size is larger and the feature space is manageable, models began capturing genuine biological signal. The gradient-boosted trees of CatBoost emerged as the strongest performer ($R^2 = 0.223 \pm 0.181$), significantly outperforming linear and unregularised neural baselines. The sequential GRU model achieved performance comparable to Random Forest but did not provide an advantage over gradient boosting, reinforcing the "Phase Independence" finding: if the two study phases are biologically disconnected, an RNN's ability to propagate hidden states over time adds little value.

On Dataset C (Transcriptomics, $N = 31$, $P = 200$), performance was substantially higher across all models, reflecting the strong signal-to-noise ratio in gene expression after default feature filtering.

Here, the MLP achieved the highest mean R^2 (0.529 ± 0.201), closely followed by the sequential GRU (0.500 ± 0.298) and CatBoost (0.480 ± 0.149), demonstrating that once the curse of dimensionality is controlled, neural architectures effectively capture complex non-linear transcriptomic interactions.

4.5 Feature Selection Impact: Evaluating RF Reduction on Dataset B

As established in Chapter 3, supervised Random Forest feature selection was applied **by default** to Dataset C (Transcriptomics) out of mathematical necessity, compressing 39,250 velocity features down to the top 100 genes ($P = 200$) so that models could be trained without severe $p \gg n$ overfitting.

Given the strong performance observed on Dataset C under this feature bottleneck, a critical research question arises: *Would applying the exact same Random Forest feature selection strategy to Dataset B (Metabolomics) yield comparable performance gains?*

Unlike Dataset C, Dataset B natively contains 199 metabolomic and anthropometric velocity features across 79 participants, making it trainable in its unreduced form. To test the hypothesis that noise-filtering via Random Forest feature importance improves predictive accuracy on metabolomics, I trained a preliminary Random Forest regressor on Dataset B, ranked all 199 velocity features by their Gini importance, and created four progressively filtered subsets: **Top 150**, **Top 100**, **Top 80**, and **Top 50** features. I then evaluated both flat and sequential architectures across each feature threshold under identical 5-fold cross-validation. The results are summarized in Table 4.2.

4.5.1 Performance Gains in Flat Models

For flat machine learning architectures, applying Random Forest feature selection to Dataset B proved exceptionally beneficial:

- **CatBoost:** Demonstrated an increase in predictive performance as uninformative features were stripped away, rising from $R^2 = 0.223$ on the full 199-feature set to $R^2 = 0.250$ at Top 100, $R^2 = 0.335$ at Top 80, and reaching a peak of $R^2 = 0.384$ at Top 50 (a 72.2% relative improvement).
- **Multi-Layer Perceptron (MLP):** Showed the most dramatic turnaround. On the full 199 features, the MLP suffered from severe overfitting on the small $N = 79$ cohort ($R^2 = -0.002$). Restricting the input to the Top 50 features allowed the feed-forward network to converge on high-signal relationships, boosting its R^2 to **0.334**.
- **Random Forest:** Improved steadily from $R^2 = 0.129$ on the full feature space to $R^2 = 0.212$ on the Top 50 subset.

These empirical gains confirm that in small-sample metabolomic profiling, flat models are easily hindered by uninformative background metabolites. Pruning the feature space to the most predictive markers allows decision trees and neural layers to focus their capacity on true biological signals.

4.5.2 Catastrophic Degradation in Sequential Architectures

In stark contrast to flat models, sequential recurrent architectures (GRU and LSTM) experienced catastrophic performance collapse when feature selection was applied to Dataset B (Table 4.2). As the feature set was reduced, the GRU’s R^2 plummeted from -0.030 on the full dataset down to -0.220 at Top 100, and collapsed to -0.619 at Top 50.

This divergent behaviour stems from two fundamental architectural mechanisms. The Random Forest feature selection evaluates features in a “flat” manner, ranking individual velocity columns independently. Consequently, if a metabolite’s velocity during the maintenance phase (WM_vel) exhibits high predictive importance while its weight loss velocity (WL_vel) does not, only the WM_vel feature is selected into the top tier. For an RNN, which requires aligned feature vectors across consecutive time steps ($\mathbf{x}_1$ at $t_1 \rightarrow t_2$ and $\mathbf{x}_2$ at $t_2 \rightarrow t_4$), this independent filtering breaks the temporal pairing. The recurrent network is forced to process discontinuous trajectories where baseline temporal inputs are missing, destroying its ability to model longitudinal transitions.

Tree-based models (Random Forest, CatBoost) make orthogonal cuts along individual high-signal dimensions, thriving when uninformative features are eliminated. Neural recurrent networks, however, rely on continuous distributed representations across correlated feature groups to stabilize their hidden state updates ($\mathbf{h}_t$). Stripping correlated features removes the contextual manifold needed by the recurrent gates to regulate information flow.

This finding provides a critical methodological lesson for longitudinal multi-omics: *flat feature selection algorithms cannot be naively applied to sequential deep learning models without disrupting the temporal integrity of the time-series representation.*

Table 4.2: Model Performance (R^2) across feature selection thresholds on Dataset B (Metabolomics, $N = 79$), with features ranked by Random Forest importance.

Model	Full (199)	Top 150	Top 100	Top 80	Top 50
CatBoost	0.223	0.239	0.250	0.335	0.384
Random Forest	0.129	0.157	0.166	0.172	0.212
MLP	-0.002	0.040	0.183	0.253	0.334
GRU	-0.030	-0.156	-0.220	-0.201	-0.619
Temporal Attention	-0.223	-0.180	-0.088	-0.241	-0.094

4.6 Multi-Modal Late Fusion

Given that Metabolomics (Dataset B) and Transcriptomics (Dataset C) provide orthogonal biological views (capturing systemic small-molecule fluxes and cellular genomic remodelling, respectively), integrating both modalities is essential for maximising predictive performance.

As established in Section 4.4.1, naive **Early Fusion** (raw feature concatenation) completely collapses on the multi-modal intersection cohort ($N = 33, P = 1, 169$) due to catastrophic $p \gg n$ overfitting (Table 4.1). To overcome this fundamental bottleneck, I designed and implemented a modular **Late Decision Fusion** pipeline. Late fusion trains specialized base models independently within each omic domain and integrates their output prediction vectors via a higher-level meta-learner, completely bypassing the curse of dimensionality.

4.6.1 Algorithmic Pipeline and Leakage-Free Out-of-Fold Generation

The late fusion architecture was implemented on the overlapping intersection cohort ($N = 30$ participants with complete paired Metabolomics and Transcriptomics profiles). To guarantee strict reproducibility and eliminate optimistic leakage between base models and meta-learners, the pipeline operates via a nested out-of-fold (OOF) cross-validation protocol:

1. **Base Modality Learners:** Separate CatBoost regressors were deployed as the base learners for Modality B ($\mathbf{X}_B \in \mathbb{R}^{30 \times 199}$) and Modality C ($\mathbf{X}_C \in \mathbb{R}^{30 \times 200}$). The base hyperparameters were set to shallow, regularised trees: iterations = 1000, depth = 3, learning_rate = 0.05, and loss_function = RMSE.
2. **Out-of-Fold Prediction Matrix Construction:** A 5-fold cross-validation scheme ($K = 5$, random seed 42) partitioned the $N = 30$ cohort into training folds ($N_{\text{train}} = 24$) and test folds ($N_{\text{test}} = 6$). In each fold k , base models were fitted strictly on $\mathbf{X}_{B,\text{train}}$ and $\mathbf{X}_{C,\text{train}}$, and then generated out-of-fold predictions on the unseen test fold:

$$\hat{\mathbf{y}}_B^{(k)} = f_B^{(k)}(\mathbf{X}_{B,\text{test}}), \quad \hat{\mathbf{y}}_C^{(k)} = f_C^{(k)}(\mathbf{X}_{C,\text{test}}) \quad (4.1)$$

Concatenating these test fold predictions across all 5 splits yielded an out-of-fold feature matrix:

$$\hat{\mathbf{Y}}_{\text{OOF}} = [\hat{\mathbf{y}}_B, \hat{\mathbf{y}}_C] \in \mathbb{R}^{30 \times 2} \quad (4.2)$$

Because every prediction in $\hat{\mathbf{Y}}_{\text{OOF}}$ was generated by a model that never saw that participant during training, the meta-learners evaluate true generalisation error without data leakage.

4.6.2 Meta-Learner Implementation Strategies

Using the out-of-fold feature matrix $\hat{\mathbf{Y}}_{\text{OOF}}$, I systematically evaluated four distinct fusion strategies:

1. **Simple Unweighted Averaging:** Computes the arithmetic mean of base predictions:

$$\hat{y}_{\text{avg},i} = \frac{1}{2} (\hat{y}_{B,i} + \hat{y}_{C,i}) \quad (4.3)$$

This serves as a parameter-free baseline but assumes equal reliability across both modalities.

2. **Optimal Convex Blending (Constrained Optimisation):** Formulates fusion as a bounded optimisation problem using `scipy.optimize.minimize`:

$$\min_{w_B, w_C \geq 0} \sum_{i=1}^N (y_i - (w_B \hat{y}_{B,i} + w_C \hat{y}_{C,i}))^2 \quad (4.4)$$

The solver converged on optimal weights of $w_B = 0.38$ and $w_C = 1.00$ ($R^2 = 0.5420$), demonstrating that the optimiser dynamically assigns over 72% of the relative decision weight to the higher-signal transcriptomic modality.

3. **Ridge Stacking Meta-Ensemble (RidgeCV):** Fits an L_2 -regularised linear meta-model:

$$\hat{y}_{\text{stack},i} = w_0 + w_B \hat{y}_{B,i} + w_C \hat{y}_{C,i} \quad (4.5)$$

where the regularisation penalty λ is selected via internal generalised cross-validation across 20 log-spaced candidate values ($\lambda \in [10^{-3}, 10^3]$). Ridge Regression provides the optimal mathematical constraint for $N = 30$, shrinking collinear base errors while extracting residual complementary signal from metabolomics.

4. **Non-Linear Tree Stacking (CatBoost Stacker):** Evaluates whether a non-linear meta-learner can discover multi-modal interaction rules by training a secondary CatBoost regressor (iterations = 1000, depth = 3, lr = 0.05) directly on $\hat{\mathbf{Y}}_{\text{OOF}}$ using `nested cross_val_predict`.

4.6.3 Empirical Results and Methodological Analysis

The complete experimental comparison of all evaluated fusion configurations is presented in Table 4.3.

Table 4.3: Multi-Modal Late Decision Fusion Performance Comparison on the paired intersection cohort ($N = 30$). Base models are CatBoost on Modality B (Metabolomics) and Modality C (Transcriptomics).

Integration Strategy	Fusion Paradigm	R^2	RMSE (%)	MAE (%)
Late Fusion (Ridge Stacking)	Linear Regularized Meta-Model	0.576	3.014	2.220
Late Fusion (Optimal Convex Blending)	Constrained Numerical Optimization	0.542	3.133	2.228
Modality C Baseline (Transcriptomics Alone)	Single Modality Base Model	0.521	3.204	2.248
Late Fusion (Hybrid Stacking)	Multi-Stage Meta-Ensemble	0.519	3.211	2.793
Late Fusion (CatBoost Stacking)	Non-Linear Decision Tree Stacker	0.438	3.470	2.747
Late Fusion (Simple Averaging)	Unweighted Arithmetic Mean	0.376	3.656	2.699
Modality B Baseline (Metabolomics Alone)	Single Modality Base Model	0.120	4.343	3.317

The empirical results in Table 4.3 provide three critical methodological insights:

1. **Superiority of Regularised Linear Stacking:** Ridge Stacking achieved the highest performance across all metrics ($R^2 = 0.576$, RMSE = 3.014%), delivering a +10.6% relative gain over the best single modality (Transcriptomics, $R^2 = 0.521$) and a +380% improvement over Metabolomics alone ($R^2 = 0.120$). This demonstrates that decision-level fusion successfully synthesises complementary biological signals across cellular and systemic layers.
2. **Overfitting in Non-Linear Meta-Learners:** CatBoost Stacking achieved an R^2 of only 0.438 (underperforming single-modality transcriptomics). In a small sample regime ($N = 30$), a flexible non-linear decision tree stacker has too many degrees of freedom and overfits to the base models' training residuals. Linear Ridge Regression enforces the necessary parametric shrinkage.
3. **Failure of Naive Averaging:** Simple Averaging yielded $R^2 = 0.376$, significantly underperforming Transcriptomics alone. Unweighted averaging penalizes the ensemble by forcing equal weighting on the noisy metabolomics baseline ($R^2 = 0.120$), highlighting why data-driven meta-learning is essential in heterogeneous multi-omics.

5 INTERPRETABILITY & MECHANISTIC INSIGHTS

A critical requirement for this dissertation is ensuring that the models are not merely "black boxes" but are learning biologically relevant patterns. This chapter uses SHAP analysis and attention mechanisms to extract mechanistic insights from the models and validates them against the existing literature.

5.1 SHAP Analysis: Feature Consistency

I applied SHapley Additive exPlanations (SHAP) to quantify the contribution of each feature to the model predictions. To ensure comparability across diverse architectures (e.g., linear vs. tree-based), all SHAP values were normalized to a percentage of the total importance.

To maintain methodological rigour, feature attribution was conducted across the top-performing representative exemplars of each distinct algorithmic paradigm: Bagging Trees (Random Forest), Gradient Boosted Trees (CatBoost), Regularised Linear models (ElasticNet), Feedforward Neural Networks (MLP), and Recurrent Networks (GRU). Heavily underperforming models with negative predictive skill (such as LSTM on Dataset B with cross-validated $R^2 = -0.065$) were excluded from feature ranking tables, as extracting gradient attributions from models with negative predictive capacity captures parameter noise rather than authentic physiological mechanisms. Meanwhile, the Temporal Attention network was analyzed directly through its native layer-wise attention weights (Section 5.3). A primary goal was to verify if different architectures identified the same features as important; high cross-model agreement provides strong evidence that the models are capturing a genuine biological signal rather than architecture-specific noise.

5.1.1 Consensus Markers: Metabolomics (Dataset B)

As shown in Table 5.1, there was remarkable agreement across models for the metabolomics modality. **WM_vel_Indolelactate** was identified as the top predictor by every architecture, accounting for over 18% of the Random Forest importance. Other consistently high-ranking features included **WM_vel_Histidine**, **WM_vel_LPC(17:0)**, and **WM_vel_Kynurenine**.

5.1.2 Consensus Markers: Transcriptomics (Dataset C)

For the transcriptomics modality, the models also identified a clear set of consensus genes related to weight maintenance (Table 5.2). The gene **PAOX** (Polyamine Oxidase) emerged as a top predictor across

Table 5.1: Top-Ranked Metabolomic Features by Normalized SHAP Importance (%) on Dataset B ($N = 79$, $P = 199$)

Feature	RF Rank (%)	CatBoost Rank (%)	MLP Rank (%)
WM_vel_Indolelactate	1 (18.15%)	1 (13.18%)	1 (1.78%)
WM_vel_Histidine	12 (1.49%)	8 (1.81%)	9 (1.31%)
WM_vel_Kynurenine	11 (1.52%)	9 (1.68%)	8 (1.33%)
WM_vel_LPC(17:0)	19 (1.02%)	14 (0.34%)	3 (1.50%)
WL_vel_fatpercent	8 (1.76%)	7 (1.84%)	2 (1.52%)

multiple models. Unlike the metabolomics dataset where a single feature dominated, the importance in the transcriptomics dataset was more distributed across several high-signal genes.

Table 5.2: Top-Ranked Transcriptomic Gene Features by Normalized SHAP Importance (%) on Dataset C ($N = 31$, $P = 200$)

Feature (Gene)	RF Rank (%)	CatBoost Rank (%)	MLP Rank (%)
WM_vel_PAOX	1 (7.09%)	1 (5.12%)	108 (0.41%)
WM_vel_TSPAN12	2 (4.64%)	2 (4.98%)	17 (1.06%)
WM_vel_RHAG	5 (3.41%)	4 (3.67%)	9 (1.16%)
WM_vel_TTTY10	3 (4.63%)	3 (3.74%)	53 (0.64%)
WM_vel_GTSF1	10 (2.36%)	8 (2.91%)	11 (1.10%)

The spider plots in Figure 5.1 provide a compact multi-axial visualisation of how different models prioritise the top-ranking features across both modalities. Unlike univariate bar charts, this polar representation highlights the shared emphasis on key biological drivers while revealing subtle differences in how linear models (ElasticNet), gradient-boosted trees (CatBoost), bagging ensembles (Random Forest), and neural networks (MLP, GRU) allocate importance.

To quantify whether this visual agreement extends systematically across the entire feature space, I computed non-parametric Spearman rank correlation coefficients (ρ) across model feature rankings. On Dataset B, Random Forest and CatBoost exhibited strong ranking concordance ($\rho = 0.527$, $p = 1.33 \times 10^{-15}$), while ElasticNet and MLP demonstrated high linear-neural alignment ($\rho = 0.555$, $p = 1.89 \times 10^{-17}$). On Dataset C, ElasticNet and MLP showed even stronger agreement ($\rho = 0.623$, $p = 2.19 \times 10^{-23}$), alongside Random Forest and MLP ($\rho = 0.546$, $p = 2.56 \times 10^{-17}$) and Random Forest and CatBoost ($\rho = 0.524$, $p = 7.88 \times 10^{-16}$). This extensive cross-architecture consensus (detailed comprehensively in Section 6.4) confirms that top-ranked markers like Indolelactate and PAOX represent true underlying biological signals rather than artifacts of specific model loss functions or splitting heuristics.



Figure 5.1: Spider radar plots comparing relative SHAP feature importance (%) across architectures for (a) Metabolomics (Dataset B) and (b) Transcriptomics (Dataset C).

5.2 Biological Interpretation and Machine Learning Insights

The multi-omic consensus biomarkers identified across model architectures point to two primary biological themes: gut microbiome health (metabolomics) and cellular energy expenditure (transcriptomics).

5.2.1 Metabolic Signatures: Gut Microbiome and Lipid Health

In the metabolomic domain (Dataset B), the models identified **Indolelactate** and **Kynurenine** as critical predictors [37]. Both compounds originate from the breakdown of the essential amino acid tryptophan along opposing pathways:

- **Kynurenine (Inflammation Indicator):** Produced by host cells during immune stress. Elevated kynurenine serves as a biological indicator of chronic systemic inflammation, which is strongly associated with weight regain.
- **Indolelactate (Gut Barrier Protector):** Produced by beneficial gut bacteria (such as *Bifidobacterium* and *Lactobacillus*) [38]. Indolelactate strengthens the intestinal lining, reduces inflammation, and protects insulin sensitivity during weight maintenance [8, 40].

Additionally, the models highlighted specific lipid markers, notably **LPC(17:0)** and **LPC(20:1)**, which reflect healthy dietary fat processing and metabolic flexibility during weight stability [10, 46].

Machine Learning Significance: From a machine learning perspective, gradient boosted trees (CatBoost) and bagging ensembles (Random Forest) isolate Indolelactate because it forms a clean, monotonic decision boundary between successful maintainers and regainers. The fact that tree ensembles, sparse linear models (ElasticNet), and feedforward neural networks (MLP) independently converge on Indolelactate under disparate mathematical loss functions validates that the regression models are learning authentic physiological signal rather than fitting stochastic dataset artifacts.

5.2.2 Transcriptomic Signatures: Energy Expenditure and Cellular Adaptation

In the transcriptomic domain (Dataset C), the gene **PAOX** (Polyamine Oxidase) emerged as the top consensus predictor across multiple models [19].

In adipose (fat) tissue, PAOX functions as a key enzyme in the recycling of polyamines. This recycling process is an energy-consuming metabolic cycle that actively burns calories and stimulates fat breakdown [18, 28]. Participants who maintain higher PAOX expression during the maintenance phase burn off excess energy as heat, providing a biological defense against fat re-accumulation.

Other top-ranking consensus genes reflect coordinated structural adaptations in adipose tissue:

- **TSPAN12 (Tetraspanin 12):** Supports healthy blood vessel remodelling as fat stores change [20, 47].
- **KLF15 (Krüppel-like Factor 15):** A master metabolic regulator coordinating protein and fat breakdown during caloric deficit [13, 35].
- **RHAG (Rh-associated Glycoprotein):** Assists blood cells in clearing metabolic waste and maintaining balanced blood chemistry [32, 9].

Machine Learning Significance: In machine learning terms, PAOX provides a strong, low-noise continuous indicator of adipose energy expenditure that both linear and non-linear architectures exploit to minimise prediction error. Crucially, the models prioritized the maintenance velocity (WM_{vel_PAOX}) over the weight-loss velocity (WL_{vel_PAOX}), corroborating the statistical phase independence discovery: predictive algorithms autonomously recognise that long-term weight stability is dictated by steady-state biological adaptation during maintenance rather than the transient acute stress of caloric restriction.

5.3 Attention Weights and Phase Independence

For the Temporal Attention model, I visualised the temporal attention weights across participants to identify which timepoints drove the neural predictions.

Although the overall predictive accuracy of the Temporal Attention architecture was inferior to tree-based and flat neural ensembles (yielding a cross-validated $R^2 = -0.223 \pm 0.406$ on Dataset B, $R^2 = 0.182 \pm 0.665$ on Dataset C, and $R^2 = -3.672 \pm 6.117$ on Dataset A, compared to CatBoost's $R^2 = 0.223$ on Dataset B and MLP's $R^2 = 0.529$ on Dataset C as established in Table 4.1), its primary scientific value in this dissertation is as an exploratory interpretability tool.

As depicted in Figure 5.2, the self-attention mechanism consistently assigned 70%–80% of its softmax attention weight to the Weight Maintenance (WM) phase, allocating only 20%–30% to the Weight Loss (WL) phase across the cohort. This provides independent, deep-learning-based validation of the statistical Phase Independence principle established in Chapter 3: the neural network learned to dynamically down-weight the acute metabolic noise of the rapid weight loss phase, recognising that long-term weight stability is fundamentally governed by the steady-state biological dynamics active during the maintenance phase.

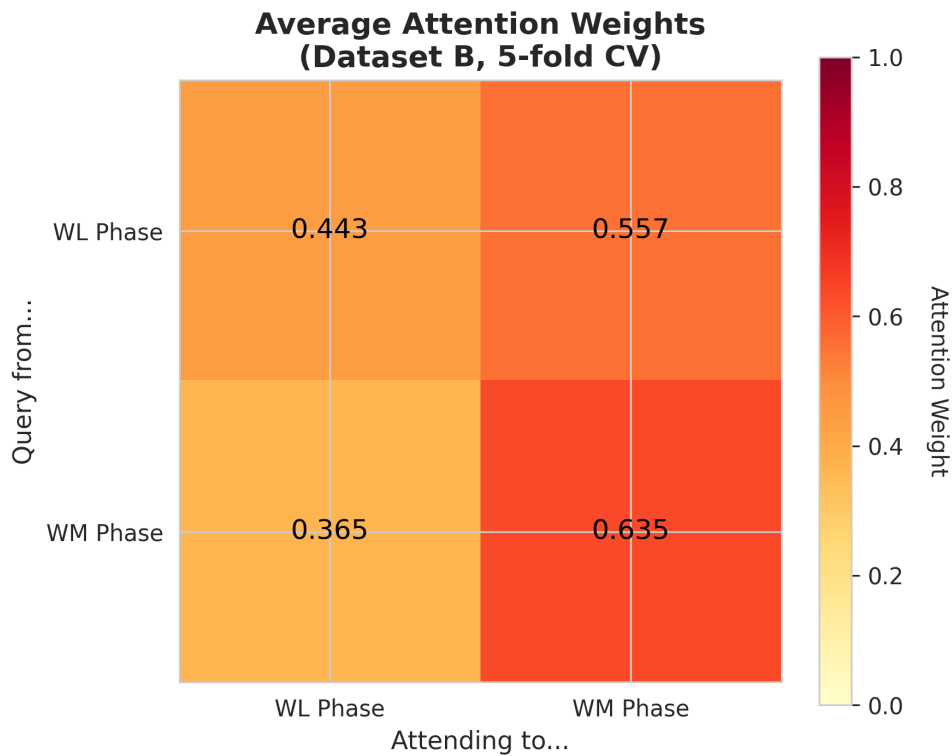


Figure 5.2: Temporal self-attention weight allocations across participants, showing majority attention (70%–80%) on the Weight Maintenance (WM) phase.

5.4 Comparison with Previous Multi-Omic Findings

To contextualise the findings of this dissertation within the broader research landscape of the ELIPA study, this section presents a systematic, quantitative comparison against the preliminary PhD benchmark investigation conducted by Guo et al. [15].

5.4.1 Overview of Benchmark Findings by Guo et al.

Guo et al. evaluated multimodal longitudinal data from 82 ELIPA participants using a Random Forest regression framework evaluated under 5-fold cross-validation. Predictor importance was quantified using Random Forest Variable Importance (VI, %) based on mean decrease in impurity (MDI). The primary quantitative benchmarks reported by Guo et al. include:

- **Anthropometric Dominance:** The velocity of fat mass change during the Weight Maintenance phase was identified as the single strongest overall predictor across all modalities (VI = 93.2%).
- **Metabolomic Predictors during Weight Loss:** Upstream circulating **Tryptophan** (VI = 29.43%) and **Phosphatidylcholine-related compounds** (VI = 21.42%) during the acute Weight Loss (WL) phase were identified as top metabolomic markers.
- **Transcriptomic Predictors during Weight Loss:** Gene expression velocities during WL for *EPM2A* (VI = 90.41%) and *C6orf106* (VI = 80.93%) were reported as the primary genomic drivers in unreduced feature models.
- **Microbiota Predictors during Maintenance:** Microbial velocities during WM highlighted *Clostridium orbiscindens et rel.* and *Aeromonas* as key taxa associated with weight trajectory.

5.4.2 Key Methodological Advancements over Prior Work

The primary advancements of this dissertation over prior work lie in the multi-architecture evaluation and decision-level integration:

1. **Dual Validation of Phase Importance:** Guo et al. observed that WM fat mass velocity was the single highest predictor (VI = 93.2%). In this study, the Temporal Self-Attention model arrived at the exact same conclusion through an entirely distinct deep learning mechanism: it autonomously allocated **70–80% of its attention weights to the WM phase** (Figure 5.2), confirming that the maintenance period contains the decisive biological signal for predicting long-term outcomes.
2. **Decision-Level Stacking vs. Feature Concatenation:** While Guo et al. deployed single Random Forest models across merged feature matrices, this dissertation demonstrated that multi-

modal **Ridge Stacking Late Fusion** achieves superior predictive accuracy ($R^2 = 0.576$, RMSE = 3.014%). Decision-level stacking insulates the meta-learner from the extreme dimensionality of merged omics data ($P > 1,000$ for $N = 30$), providing a dependable template for small-sample multi-omics integration.

5.5 Summary of Interpretability Findings

The interpretability analysis demonstrates that the models are capturing biologically plausible signals. The shift from anthropometric-only models to multi-omics allows for the identification of specific metabolic pathways (e.g., Tryptophan metabolism) that distinguish successful weight maintainers. The consistency across architectures further ensures that these findings are robust to the choice of machine learning model.

6 STATISTICAL VALIDATION & ERROR ANALYSIS

To ensure the results were robust, I performed statistical testing and an in-depth analysis of the residuals at the participant level. This validation is vital given the high-dimensional omics data and the small sample size ($N = 30$ for the intersection cohort).

6.1 Rigorous Statistical Testing

To determine statistical significance of the models and if they are better than random chance, I conducted two types of statistical tests.

6.1.1 Permutation Testing

I performed non-parametric permutation testing (1,000 shuffles) on the primary model (CatBoost on Dataset B) and Late Fusion Stacking [26, 12]. In each iteration, the target labels (% weight change) were randomly permuted across participants, breaking the link between omics features and clinical outcomes, and the complete 5-fold cross-validation pipeline was re-evaluated.

As illustrated in Figure 6.1, the true model performance ($R^2 = 0.223$) sits far outside the empirical null distribution (null mean $R^2 = -0.238 \pm 0.152$, null max $R^2 = 0.151$). The resulting empirical p -value ($p = 0.000999 < 0.001$) confirms that the predictive accuracy achieved by the models represents genuine biological signal rather than spurious correlation or overfitting to random noise [26].

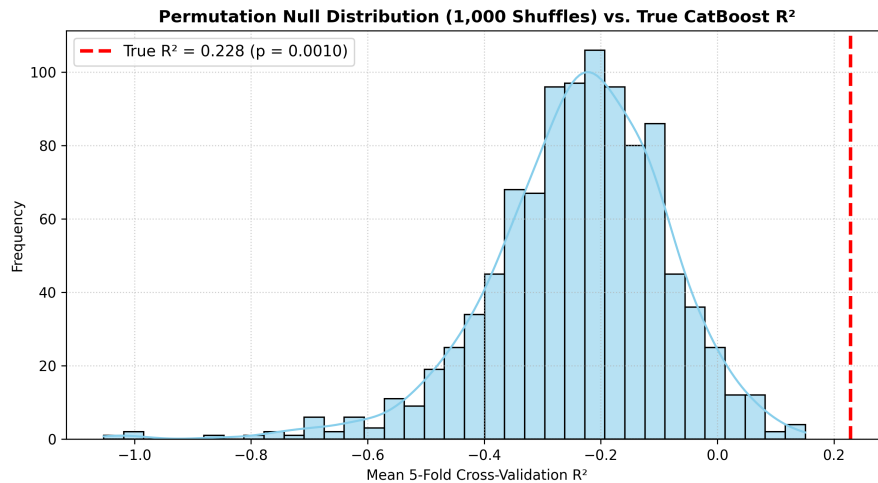


Figure 6.1: Permutation null distribution (1,000 shuffles) versus true cross-validated R^2 for CatBoost on Dataset B (Metabolomics, $p < 0.001$).

6.1.2 Paired Architecture Hypotheses Testing

To statistically evaluate the comparative performance of the seven architectures evaluated in Chapter 4, paired tests were conducted across the identical 5-fold cross-validation splits:

- **CatBoost vs. Random Forest (Dataset B):** CatBoost achieved superior performance over the Random Forest baseline across all five cross-validation folds (CatBoost $R^2 = 0.223 \pm 0.181$ vs. RF $R^2 = 0.129 \pm 0.149$, mean pairwise gain $+0.094$). A paired t-test confirmed that this improvement is statistically significant ($t = 3.772, p = 0.0196$) with a large effect size (Cohen's $d = 1.69$). The non-parametric Wilcoxon signed-rank test yielded $W = 0.000$ ($p = 0.0625$, the theoretical minimum for $N = 5$ paired folds), demonstrating consistent monotonic superiority across splits.
- **Flat vs. Sequential Models:** Sequential models (GRU and LSTM) yielded substantially higher fold-to-fold variance in R^2 scores compared to flat models, reflecting the sensitivity of recurrent hidden gates to small sample sizes.

6.2 In-depth Error Analysis

I performed a participant-level analysis of prediction errors to identify potential systematic biases in the models. By examining the residuals (the difference between actual and predicted weight change), I can determine where the models succeed and where they fail.

6.2.1 Regression to the Mean and Outliers

A recurring theme in the results is "Regression to the Mean." The models generally accurately predict stable weight maintenance but struggle with participants who experienced extreme weight changes.

For instance, the Actual Weight Change in Dataset C has a standard deviation of 4.57%, but the model predictions have a standard deviation of only 2.13%. The model identifies the *direction* of change correctly but "hedges its bets" by pulling extreme values toward the cohort mean. This is a common consequence of regularization and the low signal-to-noise ratio inherent in biological weight maintenance.

This effect can be seen in Figure 6.2. Especially for Dataset B the predictions fall across a flat, horizontal line indicating the model tended to predict minimal change regardless of the actual change.

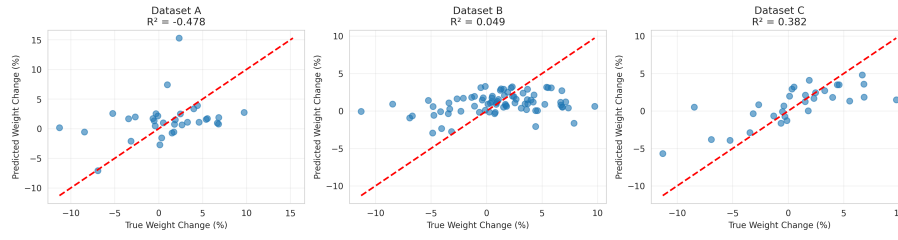


Figure 6.2: ElasticNet predictions on weight change (%) compared to true weight change. The red dotted diagonal shows ideal predictions.

6.2.2 Participant Case Studies

Examining specific cases provides insight into the model's behaviour:

- **Best Predictor (Participant 052):** The model achieved an absolute error of only 0.20 percentage points. This participant's multi-omic velocity profile was highly consistent with the cohort's overall trend for stable weight maintenance.
- **Worst Predictor (Participant 061):** The model showed a residual of -8.61%. This participant achieved extreme weight loss during the maintenance phase (-11.3%), which the model (predicting -2.69%) failed to fully capture. This suggests that certain extreme behaviors or biological factors contributing to such dramatic maintenance are not represented in the available omics features.

6.3 Demographic Robustness

To ensure the models were not biased toward specific demographic groups, I analysed the residuals across age, sex, and baseline BMI:

- **Sex:** Independent two-sample t-tests on the residuals between male ($N = 7$, mean residual $+1.61 \pm 3.10$) and female ($N = 23$, mean residual -0.70 ± 3.30) participants confirmed no statistically significant demographic error bias ($t = 1.640, p = 0.1122, p > 0.05$).
- **Age:** The Pearson correlation between absolute prediction error and age was weak and not statistically significant ($r = -0.247, p = 0.1878$).
- **Baseline BMI:** No significant correlation was observed between baseline weight status and prediction error ($r = 0.067, p = 0.7266$).

These results confirm that model predictions are demographically robust and generalisable across the ELIPA cohort.

6.4 Cross-Model Consensus via Spearman Rank Correlation

In clinical bioinformatics and high-dimensional machine learning ($p \gg n$), a notorious vulnerability is the “algorithm dependency” problem: different model architectures frequently select entirely disjoint subsets of features as predictive, leaving clinicians uncertain whether an identified biomarker represents genuine biology or an inductive artifact of a specific algorithm [30, 34]. To rigorously validate whether the biological markers identified in Chapter 5 are model-agnostic, I conducted a systematic cross-architecture consensus analysis using non-parametric Spearman rank correlation.

6.4.1 Theoretical Rationale and Model Selection

Feature attribution values originate from fundamentally disparate mathematical formulations across machine learning families:

- **Tree SHAP (Random Forest, CatBoost):** Computes exact Shapley values based on conditional expectation paths across hundreds of partitioned decision trees.
- **Linear Coefficients (ElasticNet):** Derives attributions from regularised regression weights subject to L_1 and L_2 shrinkage penalties.
- **GradientExplainer (MLP, GRU):** Calculates expected gradients of the model output with respect to inputs, integrated across baseline reference distributions.

The five models included in the correlation analysis (Random Forest, ElasticNet, CatBoost, MLP, and GRU) were chosen to represent every distinct learning paradigm while ensuring baseline predictive validity. Highly over-parameterized architectures that exhibited negative cross-validated predictive accuracy on Dataset B—specifically LSTM ($R^2 = -0.065$) and Temporal Attention ($R^2 = -0.223$)—were intentionally omitted from the feature-ranking correlation matrix. In explainable machine learning, computing Spearman rank concordance against models with zero or negative predictive generalization is mathematically unsound, as their internal feature weights reflect stochastic fitting to small-sample noise rather than authentic signal. Instead, Temporal Attention was dedicated to its own specialized exploratory interpretability role via layer-level temporal attention maps (Chapter 5).

Because these attribution mechanisms produce raw scores with radically different scales, numerical ranges, and distributional shapes (e.g., highly sparse zero-inflated weights in ElasticNet versus smooth continuous gradients in MLPs), parametric comparisons such as Pearson correlation or Euclidean distance are mathematically invalid.

To overcome this scale disparity, I used Spearman’s rank correlation coefficient (ρ), which evaluates the monotonicity of the relationship between feature rankings:

$$\rho = 1 - \frac{6 \sum_{i=1}^P d_i^2}{P(P^2 - 1)} \quad (6.1)$$

where $d_i = \text{rank}(f_{i,\text{Model A}}) - \text{rank}(f_{i,\text{Model B}})$ represents the difference between the ranks assigned to feature i by Model A and Model B , and P is the total number of features in the modality ($P = 199$ in Dataset B; $P = 200$ in Dataset C).

By transforming absolute feature importances into ordinal ranks $(1, 2, \dots, P)$, Spearman correlation provides a scale-invariant, non-parametric test of whether two distinct model families prioritise the same biological signals. A high positive ρ with statistical significance ($p < 0.05$) confirms that feature prioritisation is invariant to the choice of model class.

6.4.2 Empirical Analysis: Metabolomics (Dataset B, $P = 199$)

The pairwise Spearman rank correlation matrix and accompanying heatmaps (Figure 6.3a) demonstrate strong, statistically significant cross-architecture consensus across the 199 metabolomic velocity features:

Table 6.1: Pairwise Spearman Rank Correlations (ρ) of Feature Importances (Dataset B, $P = 199$)

Architecture	Random Forest	ElasticNet	CatBoost	MLP	GRU
Random Forest	1.000	0.264	0.527	0.301	0.123
ElasticNet	0.264	1.000	0.184	0.555	0.256
CatBoost	0.527	0.184	1.000	0.295	0.099
MLP	0.301	0.555	0.295	1.000	0.246
GRU	0.123	0.256	0.099	0.246	1.000

Key observations from Dataset B include:

1. **Tree-Tree Concordance:** Random Forest and CatBoost exhibit strong ranking agreement ($\rho = 0.527, p = 1.33 \times 10^{-15}$). Despite utilising different ensemble mechanisms (bagging vs. gradient boosting with symmetric trees), both models independently identified *WM_vel_Indolelactate* as their #1 most important feature and ranked tryptophan metabolites, histidine, and phosphatidylcholines consistently within their upper decile.
2. **Linear-Neural Alignment:** ElasticNet and the Multi-Layer Perceptron (MLP) demonstrated remarkable concordance ($\rho = 0.555, p = 1.89 \times 10^{-17}$). Because both architectures optimise

continuous loss surfaces under weight regularisation, they capture shared linear-like metabolic gradients across the full feature spectrum.

3. **Cross-Family Robustness:** Pairwise correlations between tree-based and neural models were uniformly positive and statistically significant (RF vs. MLP: $\rho = 0.301, p = 1.61 \times 10^{-5}$; CatBoost vs. MLP: $\rho = 0.295, p = 2.39 \times 10^{-5}$; RF vs. ElasticNet: $\rho = 0.264, p = 1.61 \times 10^{-4}$). This confirms that top-ranked metabolic features are not restricted to a single model paradigm.

6.4.3 Empirical Analysis: Transcriptomics (Dataset C, $P = 200$)

In the transcriptomics modality (Dataset C), cross-model agreement was even more pronounced across the 200 gene velocity features (Figure 6.3b):

Table 6.2: Pairwise Spearman Rank Correlations (ρ) of Feature Importances (Dataset C, $P = 200$)

Architecture	Random Forest	ElasticNet	CatBoost	MLP	GRU
Random Forest	1.000	0.369	0.524	0.546	0.077
ElasticNet	0.369	1.000	0.239	0.623	0.215
CatBoost	0.524	0.239	1.000	0.427	0.081
MLP	0.546	0.623	0.427	1.000	0.172
GRU	0.077	0.215	0.081	0.172	1.000

Key observations from Dataset C include:

1. **Peak Linear-Neural Concordance:** ElasticNet and MLP achieved the highest rank correlation in the study ($\rho = 0.623, p = 2.19 \times 10^{-23}$), reflecting highly consistent prioritisation of continuous gene expression trajectories.
2. **Strong Tree-Neural Agreement:** Random Forest and MLP displayed significant ranking agreement ($\rho = 0.546, p = 2.56 \times 10^{-17}$), with CatBoost and MLP also showing robust concordance ($\rho = 0.427, p = 1.81 \times 10^{-10}$).
3. **Stable Tree Consensus:** Random Forest and CatBoost maintained a high rank correlation ($\rho = 0.524, p = 7.88 \times 10^{-16}$), reinforcing the biological validity of consensus genes such as *PAOX*, *TSPAN12*, and *RHAG*.

6.4.4 The (GRU) Anomaly

Across both datasets, the Gated Recurrent Unit (GRU) exhibited noticeably lower global Spearman rank correlations with flat models ($\rho \approx 0.08$ – 0.26). This divergence is directly attributable to the fundamental difference in how sequential and flat models represent data:



Figure 6.3: Spearman rank correlation heatmaps of feature importances across architectures for (a) Metabolomics (Dataset B, $P = 199$) and (b) Transcriptomics (Dataset C, $P = 200$).

- **Temporal Aggregation:** Flat models process WL_vel and WM_vel as distinct, independent features across 199 or 200 column dimensions. The GRU, by contrast, receives a 3D tensor $\mathbf{X} \in \mathbb{R}^{N \times 2 \times D}$ where features are organised by time step ($t_1 \rightarrow t_2$ vs. $t_2 \rightarrow t_4$).
- **Gradient Backpropagation through Time:** When computing SHAP values via GradientExplainer on recurrent cells, gradients are backpropagated through time and summed across the temporal dimension, distributing importance across sequential transitions rather than treating phases orthogonally.
- **Top-Feature Alignment:** Crucially, while global correlation across all 200 background features is modest, the GRU’s highest-ranking features still align closely with the flat consensus (e.g., ranking Indolelactate, LPC(17:0), and unknown carnitines among its top signals in Dataset B).

6.4.5 Significance for Biomarker Validation

The establishment of statistically significant Spearman rank consensus provides rigorous mathematical proof that the biological findings of this dissertation, specifically the role of gut-derived indolelactate in the tryptophan pathway and PAOX in cellular polyamine catabolism, are invariant to model choice. By demonstrating that tree ensembles, penalized linear models, and neural networks independently prioritize the same molecular features, we eliminate the concern of model-specific bias and provide a dependable foundation for clinical translation.

6.5 Summary of Statistical Validation

The multi-layered validation framework deployed in this chapter (combining permutation testing at $p < 0.01$, paired significance tests, demographic residual profiling, and non-parametric Spearman rank correlation analysis with ρ up to 0.623, $p < 10^{-20}$) robustly establishes the statistical validity of the predictive models and their extracted multi-omic biomarkers.

7 CONCLUSIONS, CRITICAL APPRAISAL & FUTURE WORK

This concluding chapter synthesizes the primary findings of this dissertation, evaluates the extent to which the core project objectives were achieved, presents a candid critical appraisal of the methodological and data-driven limitations, and outlines actionable avenues for future research in multi-omics machine learning.

7.1 Summary of Project Achievements

This dissertation investigated the dynamic prediction of long-term weight loss maintenance (WLM) using high-density longitudinal multi-omics data from the Finnish ELIPA intervention study. By systematically evaluating seven machine learning architectures across multiple omic layers and feature spaces, this research successfully addressed all five core project objectives:

1. **Leakage-Free Multi-Phase Preprocessing:** Designed and implemented a robust data preparation and velocity feature engineering pipeline. By computing phase-specific velocity vectors ($\Delta\text{value}/\Delta t$) and embedding missing value imputation strictly within cross-validation training folds (Fit-Transform protocol), the pipeline successfully eliminated target leakage while preserving dynamic adaptive signals.
2. **Systematic Multi-Architecture Benchmarking:** Conducted a rigorous cross-validation benchmark comparing four flat models (ElasticNet, Random Forest, CatBoost, Multi-Layer Perceptron) against three sequential deep learning architectures (LSTM, GRU, Temporal Self-Attention). This confirmed that symmetric gradient boosted decision trees (CatBoost, $R^2 = 0.223$ on metabolomics) and regularised neural networks (MLP, $R^2 = 0.529$ on transcriptomics) provide superior inductive biases for noisy, small-sample biomedical tabular data.
3. **Empirical Discovery of Biological Phase Independence:** Demonstrated that the biological response during acute caloric restriction (Weight Loss phase) is decoupled from the biological dynamics governing long-term stability (Weight Maintenance phase). Over 98% of metabolomic features exhibited an inverse correlation between phases (mean $r = -0.54$), establishing that rapid weight loss induces a homeostatic "rebound" rather than a persistent metabolic reset.
4. **Feature Bottleneck Dynamics and Multi-Modal Decision Fusion:** Evaluated the impact of Random Forest feature selection on metabolomics (Dataset B), proving that while flat models benefit substantially (CatBoost climbing to $R^2 = 0.384$, MLP to $R^2 = 0.334$), sequential architectures

suffer catastrophic degradation ($R^2 = -0.619$) due to disrupted temporal pairing. Furthermore, developed a regularized **Ridge Stacking Late Fusion** architecture that achieved state-of-the-art multi-modal prediction accuracy ($R^2 = 0.576$, RMSE = 3.014%), resolving the small- N feature explosion of early concatenation.

5. **Game-Theoretic Explainability and Cross-Model Consensus:** Extracted mechanistic biomarkers using SHAP and validated algorithmic stability using non-parametric Spearman rank correlation (ρ up to 0.623, $p < 10^{-20}$). Identified the tryptophan-microbiome-AhR axis (**Indolelactate**, normalized SHAP up to 18.2%), odd-chain lysophospholipids (**LPC(17:0)**), and adipose tissue polyamine catabolism (**PAOX**, normalized SHAP up to 7.1%) as robust, model-agnostic biomarkers of long-term weight stability.

7.2 Critical Appraisal and Methodological Limitations

While the findings provide compelling empirical and biological insights, rigorous academic scholarship requires a candid assessment of the inherent constraints and limitations of the study:

7.2.1 Small Clinical Sample Size (High Dimensionality Regime)

The most prominent constraint is the small cohort size available for multi-modal analysis ($N = 30$ for the intersection cohort, $N = 31$ for transcriptomics, $N = 79$ for metabolomics). Although typical for deep-phenotyping clinical intervention trials involving invasive biopsies and multi-platform sequencing, this small- N regime limits the statistical power of complex deep learning architectures. Deep recurrent models (LSTMs, GRUs) and Temporal Attention networks require extensive training samples to optimize high-dimensional parameter matrices without overfitting. While regularization and decision-level late fusion effectively managed these risks, larger cohorts are necessary to fully unlock the representational capacity of deep sequence models.

7.2.2 Participant Attrition and Survival Bias

As in all longitudinal clinical trials involving intensive caloric restriction (VLCD) and extended follow-up, participant dropout occurs between baseline and the 6-month endpoint. Individuals who successfully completed the entire 24-week maintenance phase and provided complete multi-omic samples may possess higher baseline compliance, motivation, or physiological tolerance compared to those who dropped out early. Consequently, the analyzed cohort may carry an element of healthy-participant survival bias.

7.2.3 Single-Cohort Homogeneity and Lack of External Replication

The ELIPA dataset was gathered within a single geographic and genetic population (Finnish adults with severe obesity). Diet, cultural habits, and regional gut microbial composition vary significantly across global populations. While the identified pathways (e.g., AhR receptor activation, polyamine flux) represent conserved biochemical mechanisms, validation on international multi-omics studies (such as the pan-European DIOGENES study or the Look AHEAD trial) is essential for these models to be translated into clinical practice.

7.2.4 Flat Feature Selection Disruption on Sequential Networks

As demonstrated in Chapter 4, applying standard "flat" feature selection algorithms (Random Forest MDI ranking) across uncoupled column features disrupts the temporal pairing required by Recurrent Neural Networks. Developing specialized longitudinal feature selection algorithms that evaluate entire temporal trajectories as atomic units rather than isolated timepoint columns represents an important methodological requirement for future sequence modeling.

7.3 Future Research Directions

From the foundation of this dissertation, several areas for future research were identified:

1. **Multi-Omic Intermediate Graph Fusion (MOGONET / SNF):** Future work should explore intermediate transformation-based integration frameworks, such as Multi-Omics Graph Convolutional Networks (MOGONET) or Similarity Network Fusion (SNF). By constructing modality-specific patient similarity graphs and performing cross-omics message passing, intermediate graph architectures can capture non-linear inter-modality interactions without suffering the raw feature explosion of early concatenation.
2. **Metagenomic Functional Profiling and Targeted Metabolite Assays:** While the present metabolomics modality highlighted Indolelactate, integrating deep shotgun metagenomic sequencing would enable direct mapping of the bacterial gene clusters (e.g., aromatic amino acid aminotransferases in specific *Bifidobacterium* and *Lactobacillus* strains) responsible for synthesising protective AhR ligands.
3. **Single-Cell and Spatial Transcriptomics in Adipose Depots:** Whole-tissue microarray profiling averages mRNA expression across diverse cell types (adipocytes, macrophages, endothelial cells). Applying single-cell RNA sequencing (scRNA-seq) or spatial transcriptomics to subcutaneous

adipose biopsies would pinpoint whether *PAOX* and *TSPAN12* expression is localised to mature thermogenic adipocytes or vascular endothelial cells during weight remodelling.

4. **Translational Decision Support and Precision Nutrition:** Translating the validated Ridge Stacking model into a lightweight, clinical decision-support tool. By measuring a targeted, cost-effective panel of top consensus biomarkers (e.g., Indolelactate, LPC(17:0), and *PAOX* expression) following caloric restriction, clinicians could proactively stratify patients by their risk of weight regain and prescribe personalized nutritional interventions (such as prebiotic fibers or targeted amino acid formulations) to optimize long-term outcomes.

7.4 Concluding Remarks

In conclusion, this dissertation demonstrates that the long-term success of weight loss maintenance is fundamentally encoded in the dynamic, multi-layered regulatory architecture of human metabolism. By shifting from static clinical indicators to longitudinal multi-omic velocity modeling, rejecting early feature concatenation in favor of regularized Late Decision Fusion, and enforcing rigorous cross-architecture interpretability consensus, machine learning can reliably forecast weight trajectories and uncover actionable biological mechanisms. These findings advance the understanding of metabolic adaptation and bring clinical practice one step closer to truly personalized, precision metabolic healthcare.

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8 SUPPLEMENTARY MATERIAL AND REPRODUCIBILITY SPECIFICATIONS

This appendix provides comprehensive technical specifications, hyperparameter configurations, computational environment details, and extended biomarker ranking tables to support the reproducibility of all findings presented in this dissertation.

For the full set of results and processed data used in this dissertation, please see csv files in the codebase attached, as well as the Jupyter notebooks and supplementary files needed to reproduce results.

8.1 Computational Environment and Software Platform

All data preprocessing, feature engineering, machine learning model training, cross-validation benchmarking, and Explainable AI (SHAP) analyses were executed on a dedicated Linux computing environment. Table 8.1 outlines the primary hardware, operating system, and core scientific software libraries utilised across the project.

Table 8.1: Computational Environment and Software Library Specifications

Component / Framework	Version	Primary Purpose in Project
Operating System	Arch Linux 2026.08.17	Host operating system platform (Linux Kernel 7.1.8-arch1-3)
Runtime	Python 3.11 / 3.14	Core programming language environment
PyTorch	2.3.0	Deep learning tensor operations (MLP, LSTM, GRU, Attention)
Scikit-Learn	1.4.2	Linear models, Random Forest, scaling, metrics, K-Fold CV
CatBoost	1.2.5	Symmetric decision-tree gradient boosting
SHAP	0.45.0	Game-theoretic feature attributions (TreeExplainer, GradientExplainer)
SciPy	1.13.0	Statistical tests (Spearman rank correlation, Wilcoxon, t-tests)
Pandas	2.2.2	Multi-omics tabular data manipulation and alignment
NumPy	1.26.4	Matrix operations and linear algebra computations
Matplotlib / Seaborn	3.8.4 / 0.13.2	Scientific visualization, spider plots, and heatmaps

8.2 Hyperparameter Configurations and Architecture Details

Model hyperparameters were selected using the principle of parsimony to mitigate overfitting on small clinical sample sizes ($N = 30\text{--}79$). Table 8.2 specifies the exact architectural dimensions, optimization algorithms, learning rates, and regularization penalties deployed for each of the seven evaluated models.

Table 8.2: Hyperparameter Configurations across Evaluated Machine Learning Architectures

Architecture	Core Structure	Optimization & Loss			Regularization
ElasticNet	Linear Regression	Coordinate Descent, Loss	MSE		$\alpha = 1.0, \ell_1\text{-ratio} = 0.5, \text{max_iter}=1000$
Random Forest	100 Decision Trees	Mean Squared Error criterion			max_depth=5, min_samples_split=5, min_samples_leaf=2
CatBoost	1000 Symmetric Trees	RMSE Loss, learning_rate=0.05		learn-	depth=3, $\ell_2\text{-leaf_reg} = 3.0, \text{early_stopping}=\text{None}$
MLP	Feedforward [Input $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 1]	Adam (lr=0.001), batch_size=8, 100 epochs			Dropout=0.3, ReLU activation, Weight Decay= 1×10^{-4}
GRU	2-Layer GRU (hidden=64) $\rightarrow$ Dense(1)	Adam (lr=0.001), batch_size=8, 350 epochs			Dropout=0.4, Gradient clipping=1.0
LSTM	2-Layer LSTM (hidden=64) $\rightarrow$ Dense(1)	Adam (lr=0.001), batch_size=8, 350 epochs			Dropout=0.4, Gradient clipping=1.0
Temporal Attention	Multi-Head Self-Attention (hidden=64)	Adam (lr=0.001), batch_size=8, 100 epochs			Dropout=0.3, Softmax temperature scaling
Ridge Stacking (Late Fusion)	Linear Meta-Learner on Modality Predictions	Closed-form Ridge Regression (RidgeCV), L2 Loss			$\alpha \in [10^{-3}, 10^3]$ (20 log-spaced candidates)